

3.2.2 Unambiguous ranges

The discrete nature of $\mathbf{F}$ implies that the estimation result is only non-ambiguous if $|T_O f_D| < 1/2$ and $\tau \Delta f < 1$ are always true. The maximum unambiguous range is therefore

$$d_{\text{unamb}} = \frac{c_0 \tau_{\text{unamb}}}{2} = \frac{c_0}{2 \Delta f}, \quad (3.25)$$

and the maximum unambiguous velocity is

$$v_{\text{unamb}} = \frac{f_{D,\text{unamb}} c_0}{2 f_C} = \frac{c_0}{2 f_C \cdot T_O}. \quad (3.26)$$

As long as the true distances stay within $d < d_{\text{unamb}}$ and the true relative velocity stays within $|v| < v_{\text{unamb}}$, there will be no ambiguity due to aliasing. However, two targets at distances d and $d + d_{\text{unamb}}$ cannot be distinguished. If the parameters are chosen well, this is not a problem because either the unambiguous range is large enough to encompass all targets, or the scattered energy from targets beyond d_{unamb} received is too small to be detected.

3.3 Periodogram-based estimation algorithms

For the (one-dimensional) case of identifying sinusoids in a discrete-time signal, the periodogram is a well-understood tool and is in fact the optimal solution if the sinusoids are well resolved (i.e. they do not lie too closely together) [85, Chap. 13]. Given a discrete-time signal $s(k)$ of length N samples, the periodogram is defined as

$$\text{Per}_{s(k)}(f) = \frac{1}{N} \left| \sum_{k=0}^{N-1} s(k) e^{-j2\pi f k} \right|^2. \quad (3.27)$$

The common way to calculate this in digital systems is to quantize the frequency in regular intervals and use the Fast Fourier Transformation

(FFT),

$$\text{Per}_{s(k)}(n) = \frac{1}{N} \left| \sum_{k=0}^{N-1} s(k) e^{-j2\pi \frac{nk}{N_{\text{Per}}}} \right|^2 \quad (3.28)$$

$$= \frac{1}{N} |\text{FFT}_{N_{\text{Per}}} [s(k)]|^2, \quad (3.29)$$

where the notation $\text{FFT}_{N_{\text{Per}}} [s(k)]$ is used to denote an FFT of length N_{Per} on the input vector $s(k)$. Note that N_{Per} does not have to be equal to N . If $N_{\text{Per}} > N$, zero-padding is used to increase the length of $s(k)$ to N_{Per} . This increases the number of supporting points of the discrete periodogram and hence the accuracy at which frequencies can be estimated.

Since OFDM radar algorithms are sinusoidal identification algorithms (see Section 3.2), the periodogram can be applied here with some modifications, which was first proposed by Sturm et al. in [13, 12, 15] and is fully described in [18].

The input data for the periodogram is the matrix $\mathbf{F}$. Since this is a two-dimensional object, the periodogram has to be extended to two dimensions. The proposed solution is [18, 4]

$$\text{Per}_{\mathbf{F}}(n, m) = \frac{1}{NM} \left| \underbrace{\sum_{k=0}^{N_{\text{Per}}-1} \left(\sum_{l=0}^{M_{\text{Per}}-1} (\mathbf{F})_{k,l} e^{-j2\pi \frac{lm}{M_{\text{Per}}}} \right) e^{j2\pi \frac{kn}{N_{\text{Per}}}}}_{N \text{ FFTs of length } M_{\text{Per}}} \right|^2 \quad (3.30)$$

$$= \frac{1}{NM} |\text{CPer}_{\mathbf{F}}(n, m)|^2. \quad (3.31)$$

The result of the sums inside the modulus operator is called *complex periodogram* (CPer) in the following.

Sinusoids in $\mathbf{F}$ will result in a peak in $\text{Per}_{\mathbf{F}}(n, m)$. Periodogram-based algorithms therefore must first detect these peaks. Then, if $\text{Per}_{\mathbf{F}}(\hat{n}, \hat{m})$ corresponds to a peak value, $\mathbf{F}$ has a column-wise oscillation of frequency $\hat{\Omega}_d = 2\pi\hat{n}/N_{\text{Per}}$ and a row-wise oscillation of frequency $\hat{\Omega}_v = 2\pi\hat{m}/M_{\text{Per}}$. By applying (3.22) and (3.23), this corresponds to a target estimate of

$$\hat{d} = \frac{\hat{n}c_0}{2\Delta f N_{\text{Per}}} \quad \text{and} \quad \hat{v} = \frac{\hat{m}c_0}{2f_C T_O M_{\text{Per}}}. \quad (3.32)$$

To further adapt this periodogram to the radar case, the indices n and m are chosen such that

$$n = 0, \dots, N_{\text{Per}} - 1 \quad \text{and} \quad m = \lfloor \frac{-M_{\text{Per}}}{2} \rfloor, \dots, \lfloor \frac{-M_{\text{Per}}}{2} \rfloor - 1. \quad (3.33)$$

This allows for negative Doppler shifts (which correspond to targets moving away from the sensor) but not for negative distances, which have no physical meaning. Any periodogram-based estimate must therefore automatically lie within the intervals $0 \leq \hat{d} \leq d_{\text{max}}$ and $-v_{\text{rel,max}} \leq \hat{v} \leq v_{\text{rel,max}}$ introduced in Section 3.2.2.

The target's RCS can also be estimated from the periodogram if the point-scatter model (3.13) is used for the signal attenuation. Since the received power of the backscattered signal depends on the RCS, the amplitude of the periodogram at a peak location can directly be translated into an estimate of the RCS by

$$\hat{\sigma}_{\text{RCS}} = c_{\text{norm}} \frac{\text{Per}_{\mathbf{F}}(\hat{n}, \hat{m})}{P_{\text{Tx}} \cdot NM} \frac{(4\pi)^3 f_c^2 \hat{d}^4}{G c_0^2}, \quad (3.34)$$

where c_{norm} is a hardware-dependant normalization factor (simulations were configured such that $c_{\text{norm}} = 1$, but for measurements this value must be calibrated).

Fig. 3.5 shows an example of a periodogram. The axes have already been converted from bin indices to range and Doppler values. The colour of the periodogram bins corresponds to the power received from the target including the processing gain from the periodogram, as will be discussed in the next section.

To improve readability, the power axis is omitted in some cases. Whenever this is the case, the normalization is chosen identically to that in Fig. 3.5.

Fig. 3.6 shows a block diagram of an entire OFDM radar system. The individual components will be discussed in the following sections.

3.3.1 Signal-to-noise ratio and processing gain

Before going into a detailed analysis of the periodogram, the signal-to-noise ratio (SNR) must be defined. Assume a target is present which

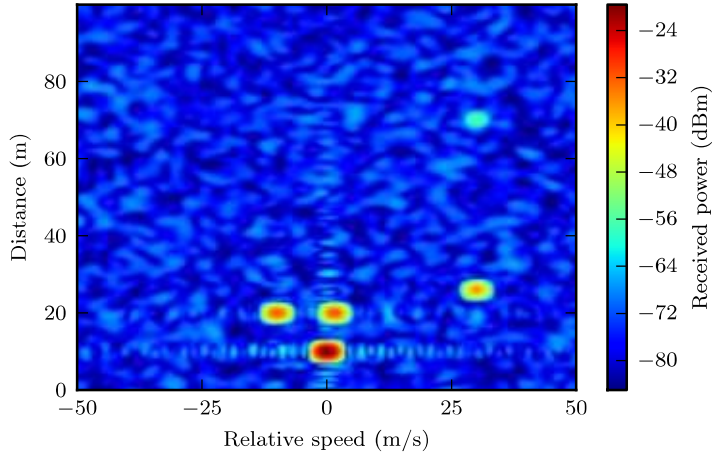


Figure 3.5: Example of a periodogram caused by five objects and WGN

backscatters a signal with power P_{Rx} . The relevant SNR is that just before the signal processing, in the matrix $\mathbf{F}_{\text{Rx}}$, which is given by

$$\text{SNR}_{\mathbf{F}_{\text{Rx}}} = \frac{P_{\text{Rx}}}{\underbrace{k_B \vartheta \cdot \text{NF}}_{\text{Noise power density}} \cdot N \Delta f}. \quad (3.35)$$

The noise power density consists of the Boltzmann constant k_B , the temperature ϑ and the noise figure NF, which incorporates the noise from all the components of the receiver chain, including the ADC. By multiplying this with the actual occupied bandwidth ($N \Delta f$), this becomes the total noise power.

Note that the division with $\mathbf{F}_{\text{Tx}}$ in (3.21) does not change the SNR, as it scales noise and received power alike. The $\mathbf{F}$ -matrix thus has the same SNR as the matrix $\mathbf{F}_{\text{Rx}}$,

$$\text{SNR}_{\mathbf{F}} = \text{SNR}_{\mathbf{F}_{\text{Rx}}}. \quad (3.36)$$

However, the target detection is performed on the periodogram, not on $\mathbf{F}$. Its signal-to-noise ratio, SNR_{Per} is therefore defined for the bin

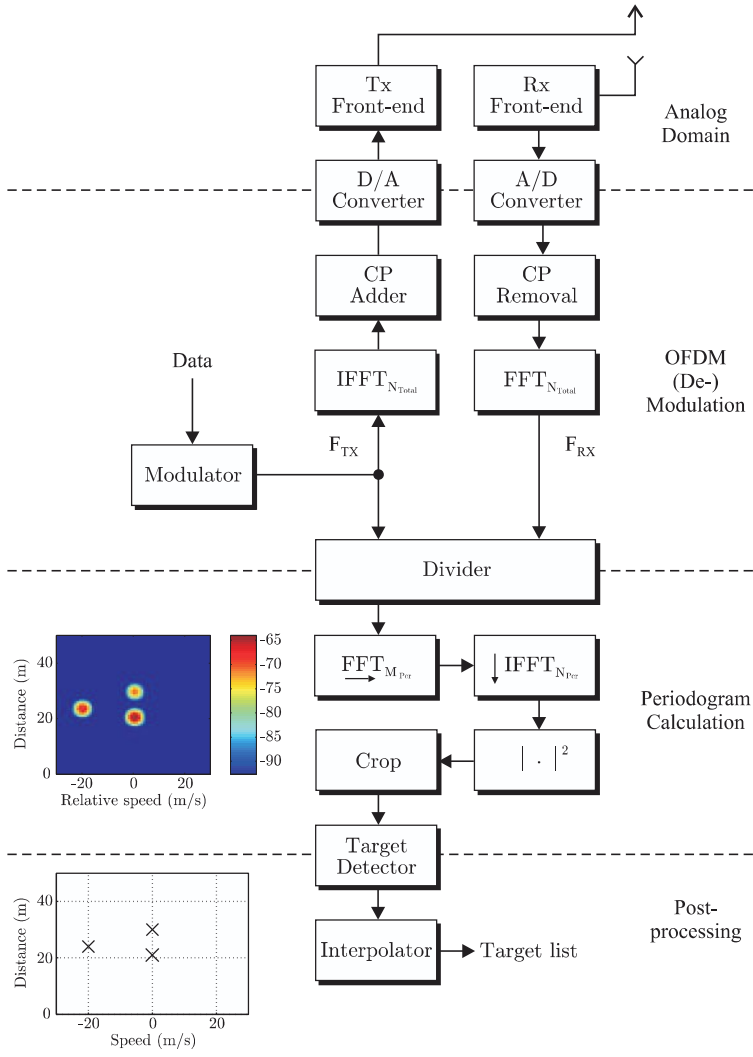


Figure 3.6: Block diagram of a periodogram-based OFDM radar system

containing the target's corresponding peak only. Intuitively, it makes sense that an optimal estimator would accumulate the received energy from all elements of $\mathbf{F}$ into one bin of the periodogram, thereby increasing the SNR by factor NM . On a logarithmic scale, this results in

$$\text{SNR}_{\text{Per}} = \text{SNR}_{\mathbf{F}} + \text{PG}, \quad (3.37)$$

where $\text{PG} = 10 \log_{10}(NM)$ is the *processing gain*² due to the periodogram.

To show this is true, consider $\mathbf{F}$ for $H = 1$ as in (3.20). First, the delay and Doppler shift are expressed by m and n ,

$$(\mathbf{F})_{k,l} = b_0 e^{j2\pi l \frac{m}{M_{\text{Per}}}} e^{-j2\pi k \frac{n}{N_{\text{Per}}}} e^{j\varphi_0} + (\mathbf{Z})_{k,l}. \quad (3.38)$$

The signal-to-noise ratio for (3.38) is

$$\text{SNR}_{\mathbf{F}} = \frac{b_0^2}{\text{E}[|(\mathbf{Z})_{k,l}|^2]}. \quad (3.39)$$

The SNR of the periodogram is, by the definition given above, the power of the signal in bin with index (n, m) divided by the noise power in that bin (note that, due to the whiteness of the noise power, the noise power is identical in all the bins):

$$\begin{aligned} \text{SNR}_{\text{Per}} = & \frac{\left| \sum_{k=0}^{N_{\text{Per}}-1} \left(\sum_{l=0}^{M_{\text{Per}}-1} b_0 e^{j2\pi l \frac{m}{M_{\text{Per}}}} e^{-j2\pi k \frac{n}{N_{\text{Per}}}} e^{j\varphi_0} e^{-j2\pi \frac{lm}{M_{\text{Per}}}} \right) e^{j2\pi \frac{kn}{N_{\text{Per}}}} \right|^2}{\text{E} \left[\left| \left(\sum_{l=0}^{M_{\text{Per}}-1} \mathbf{Z}_{k,l} e^{-j2\pi \frac{lm}{M_{\text{Per}}}} \right) e^{j2\pi \frac{kn}{N_{\text{Per}}}} \right|^2 \right]} \end{aligned} \quad (3.40)$$

$$= \frac{b_0^2 (NM)^2}{NM \cdot \text{E}[|(\mathbf{Z})_{k,l}|^2]} = NM \cdot \text{SNR}_{\mathbf{F}}. \quad (3.41)$$

As can be seen, the SNR is indeed increased by a factor NM . The numerator is a coherent accumulation of all the elements in $\mathbf{F}$, whereas the denominator is a superposition of complex Gaussian random variables.

It is important to realise that this definition of SNR_{Per} is only meaningful on a per-target basis. If several targets are present, more energy is backscattered towards the receiver—however, unless the targets are very close, this does not affect the SNR in one individual bin of the periodogram.

²In [98, Chap. 15], the general term chosen for this is *integration gain*.

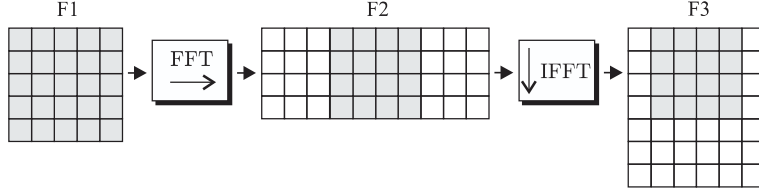


Figure 3.7: Calculation of the periodogram from $\mathbf{F}$. The grey area indicates periodogram elements within $n < N_{\max}$ and $|m| < M_{\max}$.

3.3.2 Efficient calculation of the periodogram

As indicated in (3.30), $\text{Per}_{\mathbf{F}}(n, m)$ can be calculated from $\mathbf{F}$ by means of FFTs and IFFTs. Fig. 3.7 illustrates how this is done: first, every row of $\mathbf{F}$ is processed with an FFT. From the results of these FFTs, a new matrix is created which has M_{Per} columns if zero-padding is used for the FFTs. This new matrix is then further processed by calculating the IFFT of every column. Again, zero-padding will increase the number of rows to N_{Per} . Finally, the complex values resulting from the IFFT must be turned into their magnitude-squared.

Upon further inspection, it turns out that not all $N_{\text{Per}} \times M_{\text{Per}}$ elements of the periodogram correspond to useful values. As an example, assume that there is a peak at row index $\hat{n} = N_{\text{Per}} - 1$. This would correspond to a round-trip propagation time of $\tau = \frac{N_{\text{Per}}-1}{N_{\text{Per}}}T$. Unless the guard interval T_G equals the OFDM symbol duration ($T = T_G$), this contradicts assumption (A2) (cf. Section 3.2.1)! Analogously, a peak at column index $\hat{m} = \lfloor \frac{M_{\text{Per}}-1}{2} \rfloor$ would contradict assumption (A3). If the signal parametrization is chosen correctly, there is a confined area within the periodogram with indices $n \leq N_{\max}$, $m \leq M_{\max}$ which contains all the peaks, where $N_{\max}$ and $M_{\max}$ are chosen appropriately.

The exact amount of calculations required to obtain this cropped periodogram from $\mathbf{F}$ therefore consists of:

- M IFFTs of length N_{Per} . The resulting vectors are cropped to length $N_{\max}$.

- $N_{\max}$ FFTs of length M_{Per} . Again, the resulting vectors are cropped to length $2M_{\max} + 1$.
- One magnitude-square operation for every of the remaining $N_{\max} \times (2M_{\max} + 1)$ complex values.

Calculating $\text{Per}_{\mathbf{F}}(n, m)$ thus only consists of basic signal processing steps, which can easily be implemented on FPGA- or GPP-based platforms. There are two major ways to implement this:

Serialized implementation Here, the processing is set up as a pipeline, such that the output of one step can directly be used as an input for the next, as in the block diagram in Fig. 3.6. This approach requires a very small memory footprint and is suitable for FPGA platforms which can operate at high clock rates.

Parallel implementation An alternative way is to load the entire matrix into a buffer and perform the (I)FFT operations in situ. This type of implementation is ideal for platforms which support Single-Instruction Multiple-Data (SIMD) operations, which can execute many identical signal processing operations simultaneously. With maximum parallelization, all FFTs can be executed at once (the same is true for the IFFTs), resulting in a very small execution time.

For this project, a GPP-based parallel implementation was created using the GNU Radio framework [119]. Depending on the frame size, it was possible to calculate a few hundred periodograms per second on a standard PC platform without even utilizing the entire CPU.

3.3.3 Estimation theoretic observations

The periodogram is a practical method (and, as was shown in the previous section, computationally inexpensive), but can it be justified by estimation theory as well? To answer this question, the Maximum Likelihood Estimator (MLE) shall be derived for the case where there is exactly one target and there is no prior information about the target (e.g. from a tracking system). The derivations in this section largely follow [4, 6].

A parameter vector is defined as

$$\boldsymbol{\theta} = (\tau, f_D, b, \varphi). \quad (3.42)$$

This vector also includes the phase offset, which is not required for the radar image. However, since it is unknown, it affects the estimation process, as will be seen later on.³

The derivation of a maximum likelihood estimator requires the likelihood function for $\mathbf{F}$ for a given parameter vector. As the entries of $\mathbf{F}$ are considered uncorrelated, the likelihood function is

$$f(\mathbf{F}|\boldsymbol{\theta}) = \prod_{k=0}^{N-1} \prod_{l=0}^{M-1} \frac{1}{\pi \sigma_N^2} e^{-\frac{|(\mathbf{F})_{k,l} - b e^{j(2\pi(lT_O f_D - k\tau \Delta f) + \varphi)}|^2}{\sigma_N^2}}, \quad (3.43)$$

where σ_N^2 is the variance of the elements of $\mathbf{Z}$. The maximization process is simplified when the log-likelihood function is considered,

$$\ell(\mathbf{F}|\boldsymbol{\theta}) = \sum_{k=0}^{N-1} \sum_{l=0}^{M-1} \left(-\log \pi \sigma_N^2 - \frac{1}{\sigma_N^2} \left| (\mathbf{F})_{k,l} - b e^{j(2\pi(lT_O f_D - k\tau \Delta f) + \varphi)} \right|^2 \right). \quad (3.44)$$

This function can be simplified further. The first term does not depend on any of the estimation parameters and thus does not affect maximization; it can safely be ignored. The same goes for the constant positive factor $1/\sigma_N^2$ in the second term.

The modulus squared is further evaluated using the identity $|a|^2 = a^* a$, where a^* denotes the complex conjugate:

$$\begin{aligned} \left| (\mathbf{F})_{k,l} - b e^{j(2\pi(lT_O f_D - k\tau \Delta f) + \varphi)} \right|^2 = \\ |(\mathbf{F})_{k,l}|^2 + b^2 - 2b \operatorname{Re} \left[(\mathbf{F})_{k,l} e^{-j(2\pi(lT_O f_D - k\tau \Delta f) + \varphi)} \right] \end{aligned} \quad (3.45)$$

Again, the constant terms can be ignored for maximization. The simplified log-likelihood function is thus

$$\tilde{\ell}(\mathbf{F}|\boldsymbol{\theta}) = 2b \sum_{k=0}^{N-1} \sum_{l=0}^{M-1} \operatorname{Re} \left[(\mathbf{F})_{k,l} e^{-j(2\pi(lT_O f_D - k\tau \Delta f) + \varphi)} \right] - b^2. \quad (3.46)$$

³In spectral estimation literature, it is often referred to as a *nuisance parameter*.

The maximum likelihood estimate (MLE) for the parameter vector is defined as the value which maximizes the logarithmic likelihood function [85],

$$\hat{\boldsymbol{\theta}}_{\text{ML}} = \arg \max_{\boldsymbol{\theta}} \tilde{\ell}(\mathbf{F}|\boldsymbol{\theta}). \quad (3.47)$$

The question remains as to how the MLE can be calculated in accordance with (3.47). Directly calculating the MLE from (3.46) would require solving $\nabla_{\boldsymbol{\theta}} \ell(\mathbf{F}|\hat{\boldsymbol{\theta}}_{\text{ML}}) = \mathbf{0}$ which is anything but trivial as $\hat{\boldsymbol{\theta}}_{\text{ML}}$ is only given implicitly in a non-linear set of equations.

A simpler approach can be found by rearranging (3.46):

$$\tilde{\ell}(\mathbf{F}|\boldsymbol{\theta}) = 2b \operatorname{Re} \left[e^{j\varphi} \sum_{k=0}^{N-1} \left(\sum_{l=0}^{M-1} (\mathbf{F})_{k,l} e^{-j2\pi l T_O f_D} \right) e^{j2\pi k \tau \Delta f} \right] - b^2. \quad (3.48)$$

The sum inside the real operator can be interpreted as a continuous complex periodogram. By discretizing it in the same manner as (3.29) and (3.30), a quantized log-likelihood function is given as

$$\ell_Q(\mathbf{F}|n, m, b, \varphi) = 2b \operatorname{Re} [e^{j\varphi} \text{CPer}_{\mathbf{F}}(n, m)] - b^2, \quad (3.49)$$

which is maximized w.r.t. n and m . These values can be translated into Doppler shift and delay by

$$\tau(n) = \frac{n}{N_{\text{Per}} \Delta f}, \quad (3.50)$$

$$f_D(m) = \frac{m}{M_{\text{Per}} T_O}. \quad (3.51)$$

The maximum of $\ell_Q(\mathbf{F}|n, m, b, \varphi)$ can still not be evaluated analytically, but as there is only a finite number of values for n and m , it can be solved algorithmically by searching the range of all possible values.

Finally, the phase φ has to be taken care of. If $b > 0$, then the value φ which maximizes $\ell_Q(\mathbf{F}|n, m, b, \varphi)$ is $\varphi = \arg [\text{CPer}_{\mathbf{F}}(n, m)]$. By inserting this into (3.49), $\ell_Q(\mathbf{F}|n, m, b, \varphi)$ becomes

$$\ell_Q(\mathbf{F}|n, m, b) = 2b |\text{CPer}_{\mathbf{F}}(n, m)| - b^2, \quad (3.52)$$

As $b > 0$, the values (n, m) which maximize (3.52) are those which maximize $|\text{CPer}(n, m)|$, or, equivalently, $|\text{CPer}_{\mathbf{F}}(n, m)|^2 = \text{Per}_{\mathbf{F}}(n, m)$. Finding the periodogram's peak is thus equivalent to the MLE.

Finding the value b which maximizes (3.52) can be obtained by solving $\partial/\partial b \ell_Q(\mathbf{F}|n, m, b) \stackrel{!}{=} 0$, which results in $\hat{b} = |\text{CPer}_{\mathbf{F}}(\hat{n}, \hat{m})|$. Given the point scatter model, (3.34) is thus the ML estimate for the RCS.

As discussed in the previous section, not all values (n, m) correspond to useful results. There exist values $M_{\max}, N_{\max}$ which limit the search range, which means not all $N_{\text{Per}} \times M_{\text{Per}}$ values need to be searched for a maximum. This reduces both the computational load as well as the chance of false detections.

Quantization Error

The estimates $(\hat{n}, \hat{m})$ are ML estimates, but the discretization of f_D and τ causes a quantization error. There are two ways to mitigate this: Either create a finer mesh (i.e. increase N_{Per} and M_{Per}), or find a way to locally maximize $\tilde{\ell}(\mathbf{F}|\boldsymbol{\theta})$ near $\tau(\hat{n}), f_D(\hat{m})$. Both solutions are discussed in Section 3.3.5.

Cramér-Rao Bound

The Cramér-Rao lower bound (CRB) for the frequency estimate of line spectra in discrete-time, one-dimensional processes is well known, its derivation is found in [86, 87] among others.

For a single complex sinusoid with unit amplitude in AWGN with noise power σ_N^2 and N discrete samples, the CRB for the estimate of the frequency assuming unknown phase is

$$\text{var}[\hat{\omega}] \geq \frac{6\sigma_N^2}{(N^2 - 1)N}. \quad (3.53)$$

To transfer this to the case of distance estimation, first assume there is only one OFDM symbol available ($M = 1$). It consists of N values, and using (3.32), (3.53) can directly be converted into a CRB for the distance estimate:

$$\text{var}\{\hat{d}\} \geq \frac{6\sigma_N^2}{(N^2 - 1)N} \left(\frac{c_0}{4\pi\Delta f} \right)^2. \quad (3.54)$$

Calculating the CRB for the entire frame is highly complex due to the fact that the matrix $\mathbf{F}$ consists of M OFDM symbols, each with a different, random and unknown initial phase, due to the unknown Doppler shift. Here, use is made of the fact that the presented estimators have some kind of implicit averaging to identify a simpler lower bound: First, realize that every OFDM symbol can be used for one estimation $d_i, i = 1 \dots M$. As *white* noise is postulated as the source of error, the d_i represent *independent* estimates of d . Probability theory dictates that by averaging, an estimate $\hat{d}$ is obtained with variance

$$\text{var}[d] = \frac{1}{M} \text{var}\{d_i\}. \quad (3.55)$$

This is now applied to (3.54), yielding

$$\text{var}[\hat{d}] \geq \frac{6\sigma_N^2}{(N^2 - 1)NM} \left(\frac{c_0}{4\pi\Delta f} \right)^2. \quad (3.56)$$

This bound is called the *averaged* CRB because it is not a true CRB any more, but is still a useful lower bound for the analyses presented here.

In a similar fashion, a lower bound for $\hat{v}_{\text{rel}}$ can be given,

$$\text{var}[\hat{v}_{\text{rel}}] \geq \frac{6\sigma_N^2}{(M^2 - 1)MN} \left(\frac{c_0}{4\pi T_O f_C} \right)^2. \quad (3.57)$$

It is worth pointing out the influence of SNR on the bounds, since they both share dependencies. By using $\text{SNR} = 1/\sigma_N^2$ and $B = N\Delta f$, insert (3.13) into (3.56) and (3.57), yielding

$$\text{var}[\hat{d}] \geq \frac{6(4\pi)N_0}{P_{\text{Tx}}G} \cdot \frac{r^4}{\sigma_{\text{RCS}}} \cdot \frac{f_C^2}{(N^2 - 1)M\Delta f}, \quad (3.58)$$

$$\text{var}[\hat{v}] \geq \underbrace{\frac{6(4\pi)N_0}{P_{\text{Tx}}G}}_{\text{Hardware}} \cdot \underbrace{\frac{r^4}{\sigma_{\text{RCS}}}}_{\text{Target}} \cdot \underbrace{\frac{\Delta f}{(M^2 - 1)MT_O^2}}_{\text{Signal parameters}}. \quad (3.59)$$

Interpreting (3.58) and (3.59) gives three insights in particular into the system design:

- Given the point-scatter approximation, f_C only influences $\hat{d}$ in a manner that lowering the frequency decreases the lower bound, whereas $\hat{v}_{\text{rel}}$ is unaffected.

- Increasing M decreases the lower bound for both estimates, but as this means increasing the signal duration, it also increases medium access and allows for fewer measurements per time unit.
- When increasing the bandwidth, it is advantageous to increase N rather than Δf . This is only possible within limits given by the channel characteristics, most importantly the channel's coherence bandwidth (see also Section 3.6 and [1]).

Finally, it must not be forgotten that the CRB is a suitable quality metric only for *unbiased* estimators, and comparing to the actual estimates only makes sense above a certain SNR as will be discussed in the following.

Estimation error

In [87], Rife argues that below a certain SNR (the *SNR threshold*), any point in the periodogram has an equal chance of being the largest due to the whiteness of AWGN. In OFDM radar, this is the same: When the energy backscattered by the target is too small, it becomes indistinguishable from noise in the periodogram, and any point can be the largest.

Already, this reveals several things about the estimator: At low SNR, it cannot be unbiased, and it can potentially “detect” targets which are in reality artefacts of noise (false alarms). This motivates the introduction of a threshold test to the estimation process, which involves comparing the amplitude of the detected target with a predetermined threshold to verify it as a valid target. Section 3.3.6 discusses this in greater detail.

The metric to test the performance of the MLE is thus the probability with which the estimates $(\hat{n}, \hat{m})$ are correct. Let (n_0, m_0) be the true

values, and define a “root periodogram”,

$$\begin{aligned}
 C(n, m) &:= \sqrt{\text{Per}_{\mathbf{F}}(n, m)} \\
 &= \frac{1}{\sqrt{NM}} \left| \sum_{k=0}^{N_{\text{Per}}-1} \left(\sum_{l=0}^{M_{\text{Per}}-1} (\mathbf{F})_{k,l} e^{-j2\pi \frac{lm}{M_{\text{Per}}}} \right) e^{j2\pi \frac{kn}{N_{\text{Per}}}} \right| \\
 &= \left| \underbrace{\sum_{k=0}^{N_{\text{Per}}-1} \left(\sum_{l=0}^{M_{\text{Per}}-1} e^{-j2\pi l \frac{m-m_0}{M_{\text{Per}}}} \right)}_{\text{Signal term } A(n, m)} e^{j2\pi k \frac{n-n_0}{N_{\text{Per}}}} + (\mathbf{Z})_{m,n} \right|. \quad (3.60)
 \end{aligned}$$

The signal term is given by

$$A(n, m) = b\sqrt{NM} \left| \frac{\sin(\pi N \frac{n-n_0}{N_{\text{Per}}})}{N \sin(\pi \frac{n-n_0}{N_{\text{Per}}})} \frac{\sin(\pi M \frac{m-m_0}{M_{\text{Per}}})}{M \sin(\pi \frac{m-m_0}{M_{\text{Per}}})} \right|. \quad (3.61)$$

As discussed in Section 3.3.1, the summation causes the amplitude to be scaled by $\sqrt{NM}$, whereas the noise term is still AWGN with noise power σ_N^2 , the same as on the input.

Every value $C(n, m)$ is thus a random variable with a Rician distribution of the form

$$f_{C(n, m)}(x) = \frac{2x}{\sigma_N^2} e^{-\frac{(x^2 + A^2(n, m))}{\sigma_N^2}} I_0 \left(\frac{2xA(n, m)}{\sigma_N^2} \right), \quad (3.62)$$

where $x \geq 0$ and $I_0(x)$ denotes the modified Bessel function of order zero.

The estimator result is also a random variable and shall be denoted by $(\hat{\mathbf{n}}, \hat{\mathbf{m}})$. Its PDF is given by $f_{(\hat{\mathbf{n}}, \hat{\mathbf{m}})}(n, m|n_0, m_0)$ and depends on (n_0, m_0) . From all (n, m) within the search range, the estimator decides for the particular value $(\hat{n}, \hat{m})$ whenever $C(\hat{n}, \hat{m})$ is larger than any other

$C(n, m)$. The probability for this is

$$\begin{aligned}
f_{\hat{\mathbf{n}}, \hat{\mathbf{m}}}(\hat{n}, \hat{m}|n_0, m_0) &= \Pr[C(\hat{n}, \hat{m}) > C(n, m) | \forall n \neq \hat{n}, m \neq \hat{m}] \\
&= \int_x \Pr[C(n, m) < x | n \neq \hat{n}, m \neq \hat{m}] f_{C(\hat{n}, \hat{m})}(x) dx \\
&= \int_0^\infty f_{C(\hat{n}, \hat{m})}(x) \left[\prod_{\substack{n \neq \hat{n} \\ m \neq \hat{m}}} \int_0^x f_{C(n, m)}(y) dy \right] dx \\
&= \int_0^\infty f_{C(\hat{n}, \hat{m})}(x) \left[\prod_{\substack{n \neq \hat{n} \\ m \neq \hat{m}}} F_{C(n, m)}(x) \right] dx
\end{aligned} \tag{3.63}$$

where $F_{C(n, m)}(x)$ is the cumulative density of the value of $C(n, m)$,

$$F_{C(n, m)}(x) = 1 - Q_1\left(\frac{A(n, m)\sqrt{2}}{\sigma_N}, \frac{x\sqrt{2}}{\sigma_N}\right) \tag{3.64}$$

and $Q_1(\cdot, \cdot)$ is the Marcum Q-function.

The complete PDF for $(\hat{\mathbf{n}}, \hat{\mathbf{m}})$ is calculated for a given (n_0, m_0) and SNR by solving the integral for every possible value of $(\hat{n}, \hat{m})$ in the search range.⁴

Having calculated the PDF $f_{\hat{\mathbf{n}}, \hat{\mathbf{m}}}(\hat{n}, \hat{m}|n_0, m_0)$, it is now possible to calculate the expected value and variance of the estimator. Using these and (3.32) to translate $\hat{n}$ and $\hat{m}$ into a range and velocity, respectively,

⁴(3.63) can be solved numerically, but requires special attention due to the combination of Bessel and exponential functions, which can have extremely large or small values and can thus become numerically unstable. Numerically demanding calculations were done with the help of the arbitrary precision tools for the Python programming language. The methods used to calculate $Q_1(\cdot, \cdot)$ are those explained in [70].

equations for bias and variance of the range and Doppler are

$$\text{bias}[d] = \frac{c}{2N_{\text{Per}}\Delta f} (\mathbb{E}\{\hat{\mathbf{n}}\} - n_0), \quad (3.65)$$

$$\text{var}[d] = \left(\frac{c}{2N_{\text{Per}}\Delta f} \right)^2 \mathbb{E}\{(\hat{\mathbf{n}} - \mathbb{E}\{\hat{\mathbf{n}}\})^2\}, \quad (3.66)$$

$$\text{bias}[v_{\text{rel}}] = \frac{c}{2M_{\text{Per}}T_{\text{Ofc}}} (\mathbb{E}\{\hat{\mathbf{m}}\} - m_0), \quad (3.67)$$

$$\text{var}[v_{\text{rel}}] = \left(\frac{c}{2M_{\text{Per}}T_{\text{Ofc}}} \right)^2 \mathbb{E}\{(\hat{\mathbf{m}} - \mathbb{E}\{\hat{\mathbf{m}}\})^2\}. \quad (3.68)$$

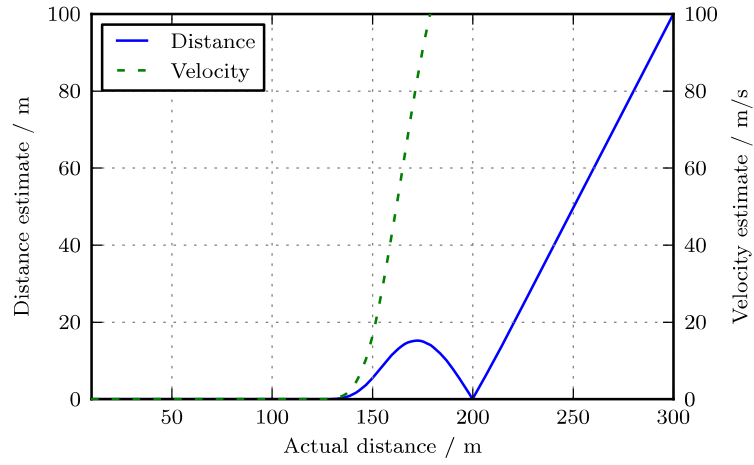
Fig. 3.8 shows bias and variance for an OFDM radar system parametrized as described in Section 3.6.2 (wideband signal). The target RCS was fixed to $\sigma_{\text{RCS}} = 10 \text{ m}^2$, which means the SNR is directly proportional to $\propto 1/d^4$. It can be seen that bias and variance stay close to zero until a certain range is exceeded (i.e. SNR falls below a certain value). As soon as this occurs, the estimation error rises very quickly, making the estimator unusable in practice beyond this SNR threshold.

Fig. 3.9 shows the probability $\Pr[\hat{n} = n_0, \hat{m} = m_0]$ of estimating the correct values. The results confirm that the estimator works very well above the SNR threshold, but very poorly below.

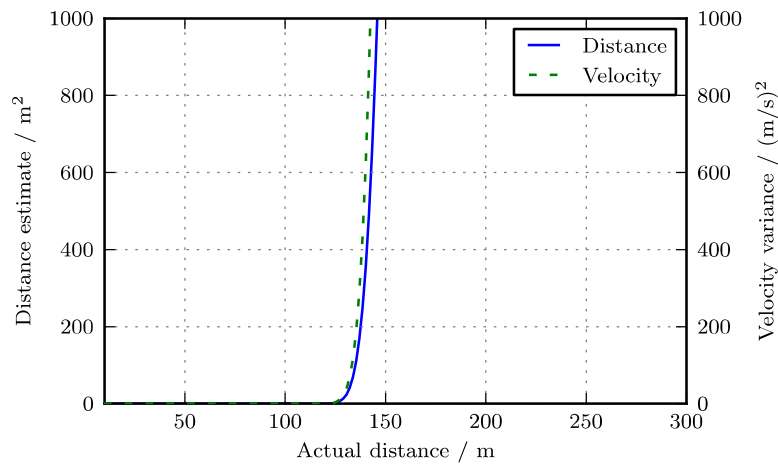
It is emphasized that (3.65) through (3.68) are calculated from an estimator for (n, m) , and not (d, v_{rel}) ! Because of the aforementioned quantization error, the range and Doppler estimates are worse than the estimates for n and m . The value of this analysis is that it provides a way to numerically calculate bias and variance of an otherwise mathematically intractable problem.

To obtain true bias and variance of a complete OFDM radar system, including quantization error and interpolation algorithms to remove the latter, simulations are required which are presented in Section 5.2.

It can be argued that the bias is not a useful metric in this case, as it does not show a systematic error of the estimation, but rather a random offset. A large bias is thus more of an indicator that the estimation is not working correctly, the numeric value of the bias itself does not hold any useful information about the estimation process. For this reason, the probability of estimating the correct bin can be chosen to determine the quality of the estimator, which is less misleading.



(a) Bias of the distance and velocity estimates.



(b) Variance of the distance and velocity estimates.

Figure 3.8: Performance of target velocity and distance estimation against actual distance

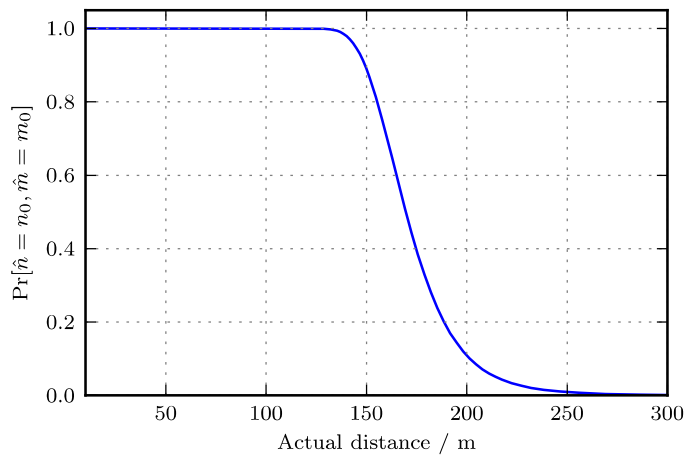


Figure 3.9: Probability of correct value estimation.

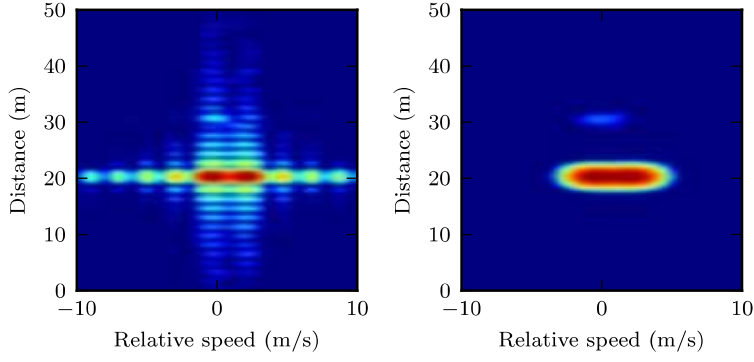
3.3.4 Window functions

In periodogram-based spectral estimation, window functions⁵ are commonly applied to the input data. Window functions have two relevant properties: the width of the main lobe Ω_s , where $-\pi \leq \Omega_s \leq \pi$, and the sidelobe attenuation $a_{\min}$, given in dB.

The main advantage is the precise control of sidelobe levels, which is useful for the reliable detection of targets. Fig. 3.10 shows the same periodogram for two identical cases, but each using a different window. In the first case, the main lobe width is very small and close targets can be distinguished (i.e. the resulting radar resolution is better). The second periodogram uses a window with high sidelobe attenuation; this allows weaker targets to be distinguished from sidelobes of the stronger targets, but reduces the resolution.

For the two-dimensional periodogram, a two-dimensional window matrix $\mathbf{W} \in \mathbb{R}^{N \times M}$ is defined, which is multiplied element-wise with the matrix

⁵Sometimes referred to as *taper functions*.



(a) Rectangular window. The two targets at 20 m are easily distinguished. (b) Dolph-Chebyshev window. The target at 30 m now clearly stands out.

Figure 3.10: Periodogram of three targets.

F. (3.30) is thus refined to

$$\text{Per}_{\mathbf{F}}(n, m) = \frac{1}{NM} \left| \underbrace{\sum_{k=0}^{N_{\text{Per}}-1} \left(\sum_{l=0}^{M_{\text{Per}}-1} (\mathbf{F})_{k,l} (\mathbf{W})_{k,l} e^{-j2\pi \frac{lm}{M_{\text{Per}}}} \right)}_{M_{\text{Per}} \text{ IFFTs of length } N_{\text{Per}}} e^{j2\pi \frac{kn}{N_{\text{Per}}}} \right|^2. \quad (3.69)$$

If $\mathbf{W}$ is left out, as in (3.30), a boxcar window $((\mathbf{W})_{k,l} = 1)$ is implicitly used.

Window matrices are designed by calculating the outer product of two one-dimensional windows,

$$\mathbf{W} = \frac{1}{\|\mathbf{w}_d\|^2 \|\mathbf{w}_v\|^2} \mathbf{w}_d^T \otimes \mathbf{w}_v, \quad \mathbf{w}_d \in \mathbb{R}^{1 \times N}, \quad \mathbf{w}_v \in \mathbb{R}^{1 \times M}. \quad (3.70)$$

The normalization factor is introduced to ensure that the total energy in the periodogram is independent of the chosen window matrix.

It is possible to choose different window types for $\mathbf{w}_d$ and $\mathbf{w}_v$, but this would only be useful for some rare cases, e.g. if there is a large difference

Window	Main lobe width	Sidelobe attenuation
Rectangular	$\Omega_s = 2\pi/N$	$a_{\min} = 13.3 \text{ dB}$
Hamming	$\Omega_s = 4\pi/N$	$a_{\min} \approx 42 \text{ dB}^7$
Blackman-Harris	$\Omega_s \approx 8\pi/N$	$a_{\min} = 92.1 \text{ dB}$
Dolph-Chebyshev	$\Omega_s \approx \frac{1.46\pi(\log_{10} 2 + a_{\min}/20)}{N-1}$	user-defined

Table 3.2: Properties of window functions used.

between the values of N and M . Here, only window matrices with identical window types for rows and columns are considered (for the sake of readability, only the definition for $\mathbf{w}_d$ is given; $\mathbf{w}_v$ can be calculated in an analog fashion by replacing N with M). The following window types are of interest for the OFDM radar application:⁶

Rectangular (boxcar) windows This can also be interpreted as a “lack” of a window function as all entries are of unit value,

$$(\mathbf{w}_d)_k = 1, \quad k = 0, \dots, N-1. \quad (3.71)$$

The rectangular window has the smallest main lobe width (which is optimal for the radar resolution). However, it also has the smallest side lobe attenuation (which is bad for the dynamic range), which makes it generally not a good choice for the radar system.

Hamming windows Compared to the rectangular window, these windows have a higher sidelobe attenuation (see Table 3.2), but twice the main lobe width. Hamming windows have been used in combination with radar systems before (e.g. in [38]) and therefore were analyzed here as well. They are defined by

$$(\mathbf{w}_d)_k = 0.54 - 0.46 \cos\left(\frac{2\pi k}{N-1}\right). \quad (3.72)$$

⁶For more information on these windows, cf. [102, Chap. 7], and [101] for details on the Blackman-Harris window.

⁷This value is only correct for $N \geq 32$.

Blackman-Harris windows An alternative to the Hamming window with an extremely high sidelobe attenuation is the Blackman-Harris window, which is defined as

$$(\mathbf{w}_d)_k = a_0 - a_1 \cos\left(\frac{2\pi k}{N-1}\right) - a_2 \cos\left(\frac{4\pi k}{N-1}\right) - a_3 \cos\left(\frac{6\pi k}{N-1}\right)$$

$$a_0 = 0.35875, a_1 = 0.48829, a_2 = 0.14128, a_3 = 0.01168. \quad (3.73)$$

The main lobe is nearly twice as wide as that of the Hamming window, but the very high sidelobe attenuation effectively allows sidelobes to be treated as if they were non-existent, because in most cases the dynamic range will actually be smaller than $a_{\min}$.

Dolph-Chebyshev windows Dolph-Chebyshev windows have a unique property: the sidelobe attenuation is constant over the entire frequency band. It is also parametrizable, which means the sidelobe attenuation can be configured according to the required dynamic range. The main lobe width depends on the chosen sidelobe attenuation (see Table 3.2). Because the sidelobe energy is constant outside the main lobe and does not decrease with Ω as with the other windows, the main lobes of Dolph-Chebyshev windows are usually slightly smaller than those of the other windows at equivalent sidelobe attenuations (e.g. a Dolph-Chebyshev window with 92.1 dB sidelobe attenuation has a slightly smaller main lobe than a Blackman-Harris window of equal length).

Dolph-Chebyshev windows are designed in the frequency domain, the final window is calculated with an Inverse Discrete Fourier Transformation (IDFT) (or Inverse Fast Fourier Transformation (IFFT)):

$$\mathbf{w}_d = \text{IFFT}_N \left[\mathbf{W}_d e^{-j(N-1)\Omega/2} \right], \quad (3.74)$$

where

$$(\mathbf{W}_d)_k = \begin{cases} \cosh\left((N-1) \cdot \text{arccosh}\left(\frac{\cos(\frac{\pi k}{N})}{\cos(\Omega_s/2)}\right)\right), & 0 \leq \Omega \leq \Omega_s, \\ \cos\left((N-1) \cdot \arccos\left(\frac{\cos(\frac{\pi k}{N})}{\cos(\Omega_s/2)}\right)\right), & \Omega_s \leq \Omega \leq \pi. \end{cases} \quad (3.75)$$

SNR loss by windowing

The application of a window does not affect the signal-to-noise ratio at the input of the periodogram, because the element-wise multiplication with $\mathbf{W}$ affects noise and signal in the same manner. The choice of the window function has an impact on the signal-to-noise ratio in the periodogram, though. Because the main lobe's widths differ between the presented windows, the height of their peaks also varies, resulting in an SNR loss.

Fig. 3.19 shows several windows both in time- and frequency domain. The SNR loss can be seen in the frequency domain; the rectangular window has the highest peak.

Such a loss needs to be factored into the processing gain. For arbitrary windows, it is therefore calculated as

$$\text{PG} = 10 \cdot \log_{10}(NM) + \text{SNR}_{\mathbf{w}_d} + \text{SNR}_{\mathbf{w}_v}, \quad (3.76)$$

where $\text{SNR}_{\mathbf{w}_d}$ and $\text{SNR}_{\mathbf{w}_v}$ are the SNR losses for the row- and column-wise windows, respectively. They are given by

$$\text{SNR}_{\mathbf{w}_d} = \frac{1}{\|\mathbf{w}_d\|^2 N} \left| \sum_{k=0}^{N-1} (\mathbf{w}_d)_k \right|^2, \quad (3.77)$$

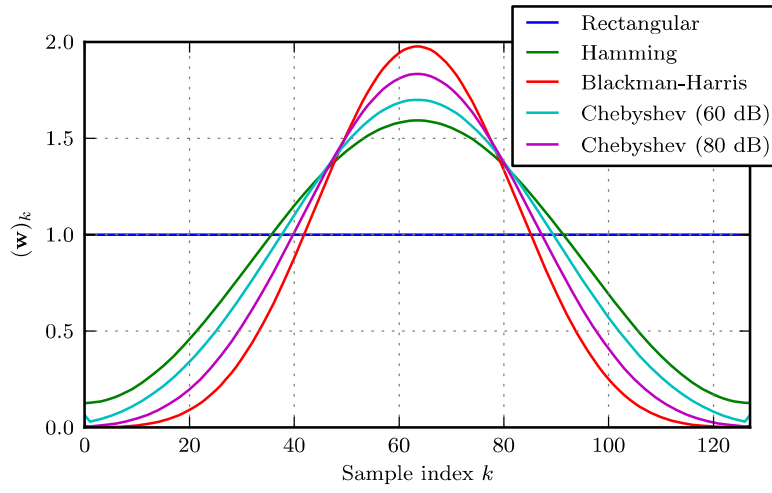
$$\text{SNR}_{\mathbf{w}_v} = \frac{1}{\|\mathbf{w}_v\|^2 M} \left| \sum_{k=0}^{M-1} (\mathbf{w}_v)_k \right|^2, \quad (3.78)$$

$$(3.79)$$

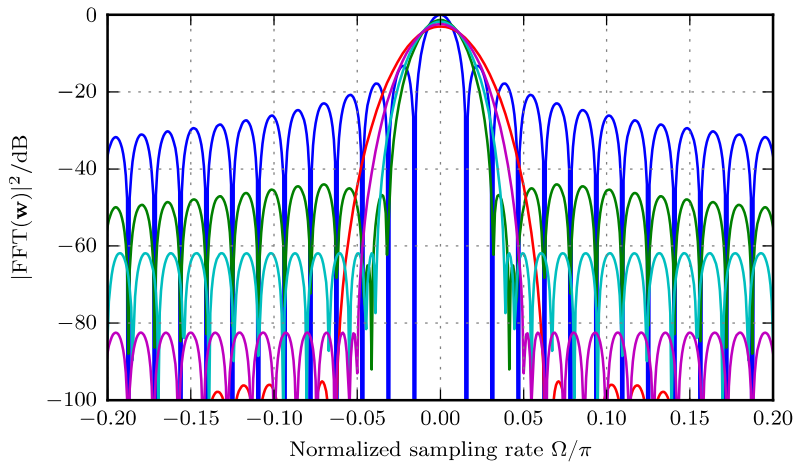
which corresponds to the DC bin of the individual windows' periodogram normalized to the height of the rectangular window. Table 3.3 gives some values for different types of windows and lengths. It should be noted that some windows induce a significant loss of SNR.

3.3.5 Quantization issues and interpolation

Up until now, the range of possible estimates is quantized on a regular grid, as the values $\hat{n}$ and $\hat{m}$ in (3.32) are integer values. This introduces two negative effects: quantization noise and scalloping loss. These are discussed in the following, as well as a solution for the quantization problem.



(a) Time-domain representation. Windows are normalized to unit energy.



(b) Frequency-domain representation (logarithmic). Windows are normalized to peak of rectangular window.

Figure 3.11: Different window functions in one dimension with length $N = 128$.

	Rectangular	Hamming	Blackman-Harris	Dolph-Chebyshev
$N = 52$	0 dB	-1.40 dB	-3.10 dB	-1.88 dB
$N = 256$	0 dB	-1.36 dB	-3.04 dB	-1.82 dB
$N = 1024$	0 dB	-1.35 dB	-3.02 dB	-1.81 dB

Table 3.3: SNR loss for different window types and -lengths

Quantization noise

On a quantized grid, estimates are only correct up to a quantization error. Assuming a uniform distribution of true range and Doppler shifts, the estimation is distorted by zero-mean, uniformly distributed quantization noise with variances (or quantization noise power)

$$\text{var}_Q[\hat{d}] = \frac{1}{12} \left(\frac{c_0}{2\Delta f N_{\text{Per}}} \right)^2, \quad (3.80)$$

$$\text{var}_Q[\hat{v}] = \frac{1}{12} \left(\frac{c_0}{2f_C T_O M_{\text{Per}}} \right)^2. \quad (3.81)$$

Not only does this mean that the estimates are incorrect, but also that the error on the estimates is non-Gaussian, and possibly not uncorrelated between measurements. This can affect some post-processing components, such as tracking algorithms, if they assume a Gaussian distribution of the estimation error, such as Kalman filters do (cf. also [98, Chap. 19]).

Scalloping loss

Scalloping loss⁸ is another effect caused by the regular grid of FFT-based processing.⁹ If the true frequency of a sinusoid does not lie directly on one of the DFT bins, not only is the frequency estimate incorrect (which distorts the range or the Doppler estimation), but also its amplitude. In the worst case, this can cause a valid target to be mistaken for noise, but even if it is detected, the estimated amplitude is probably too low, thereby increasing the error of the Radar Cross Section (RCS).

⁸In radar literature [98], this effect is usually referred to as *straddle loss*.

⁹R. Lyons [108] gives an excellent introduction to this topic.

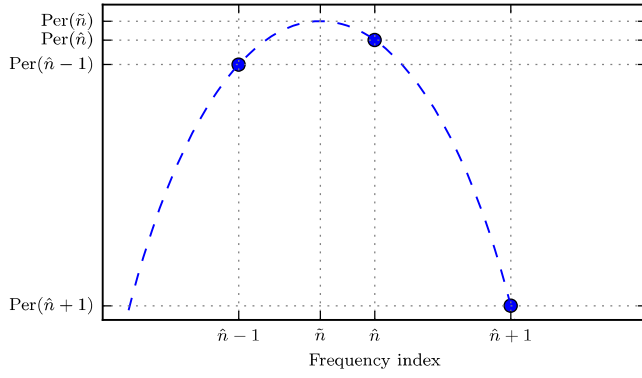


Figure 3.12: Part of the periodogram of a sinusoid with a frequency between the discrete DFT bins, normalized to its maximum value. If the maximum value of the discrete periodogram is used, the estimate will have a quantization error of $\hat{n} - \tilde{n}$ and a scalloping loss of $\text{Per}(\hat{n}) - \text{Per}(\tilde{n})$.

Fig. 3.12 shows the periodogram of a one-dimensional sinusoid, multiplied with a rectangular window. The maximum value of the continuous periodogram is between two bins, thereby causing scalloping loss and quantization noise if the FFT's peak is used as an estimate.

Interpolation

To overcome these problems, the estimation can be augmented by interpolation. One way to interpolate is to simply increase N_{Per} and M_{Per} , i.e. to pad with more zeros. While this does not change the fact that the quantization noise is uniformly distributed, its power can be decreased until the receiver noise dominates the estimation error. However, this method is not very efficient, as it interpolates the *entire* periodogram and thereby increases memory usage and computational cost.

A more efficient way is to use a periodogram with little or no zero padding to first get a coarse, quantized estimate and then interpolate

locally, which means the computational cost increases as a function of the number of detected targets.

In the following, three interpolation methods are discussed, which differ by complexity and accuracy. All three methods assume that the results from the coarse estimate are accurate within quantization, i.e. that the coarse estimate correctly identified a target (and not detected a false alarm). Also, these methods make use of the shape of the windows' main lobes, which means they work less accurately when two targets are closer to each other than the available resolution. This highlights the fact that the *accuracy* of a radar system can be improved with such methods, but the range and Doppler resolution are always fixed by the bandwidth and frame duration, respectively.

Relying on a correct initial estimate also implies that missed detections caused by scalloping loss cannot be improved using interpolation.

All interpolation methods result in *fractional* (i.e. real-valued) periodogram indices $(\tilde{n}, \tilde{m})$. These indices can be used instead of $\hat{n}$ and $\hat{m}$ in (3.32) to obtain a non-quantized estimate for range and Doppler.

Linear interpolation Here, not only $\text{Per}_{\mathbf{F}}(\hat{n}, \hat{m})$ is considered but also the four points in the periodogram directly adjacent. The periodogram must have a local maximum at $\text{Per}_{\mathbf{F}}(\hat{n}, \hat{m})$. Fractional periodogram indices $\tilde{n}$ and $\tilde{m}$ are calculated by

$$\tilde{n} = \frac{(\hat{n} - 1) \text{Per}_{\mathbf{F}}(\hat{n} - 1, \hat{m}) + \hat{n} \text{Per}_{\mathbf{F}}(\hat{n}, \hat{m}) + (\hat{n} + 1) \text{Per}_{\mathbf{F}}(\hat{n} + 1, \hat{m})}{\text{Per}_{\mathbf{F}}(\hat{n} - 1, \hat{m}) + \text{Per}_{\mathbf{F}}(\hat{n}, \hat{m}) + \text{Per}_{\mathbf{F}}(\hat{n} + 1, \hat{m})}, \quad (3.82)$$

$$\tilde{m} = \frac{(\hat{m} - 1) \text{Per}_{\mathbf{F}}(\hat{n}, \hat{m} - 1) + \hat{m} \text{Per}_{\mathbf{F}}(\hat{n}, \hat{m}) + (\hat{m} + 1) \text{Per}_{\mathbf{F}}(\hat{n}, \hat{m} + 1)}{\text{Per}_{\mathbf{F}}(\hat{n}, \hat{m} - 1) + \text{Per}_{\mathbf{F}}(\hat{n}, \hat{m}) + \text{Per}_{\mathbf{F}}(\hat{n}, \hat{m} + 1)}. \quad (3.83)$$

The linear interpolation offers no way to interpolate the height of the periodogram at $(\tilde{n}, \tilde{m})$ and is therefore as susceptible to scalloping loss as the coarse estimate.

Quadratic interpolation By approximating the main lobe of the window with a two-dimensional parabola, the estimation can be further improved. Because no window's lobe is exactly shaped like a parabola, an error still remains, but the calculation of the estimate is very simple; if $(\hat{n}, \hat{m})$ are the coordinates of a peak in the periodogram, fractional indices are calculated by

$$\tilde{n} = \hat{n} + \frac{\text{Per}_{\mathbf{F}}(\hat{n} - 1, \hat{m}) - \text{Per}_{\mathbf{F}}(\hat{n} + 1, \hat{m})}{2(\text{Per}_{\mathbf{F}}(\hat{n} - 1, \hat{m}) + \text{Per}_{\mathbf{F}}(\hat{n} + 1, \hat{m}) - 2\text{Per}_{\mathbf{F}}(\hat{n}, \hat{m}))}, \quad (3.84)$$

$$\tilde{m} = \hat{m} + \frac{\text{Per}_{\mathbf{F}}(\hat{n}, \hat{m} - 1) - \text{Per}_{\mathbf{F}}(\hat{n}, \hat{m} + 1)}{2(\text{Per}_{\mathbf{F}}(\hat{n}, \hat{m} - 1) + \text{Per}_{\mathbf{F}}(\hat{n}, \hat{m} + 1) - 2\text{Per}_{\mathbf{F}}(\hat{n}, \hat{m}))}. \quad (3.85)$$

As before, the interpolated values for the range and Doppler estimation are then calculated by replacing the integer values with the fractional values $\tilde{m}$ and $\tilde{n}$ in (3.32), respectively.

Another advantage is that the interpolated value of the parabola at position $(\tilde{n}, \tilde{m})$ can be used for the RCS estimation, thereby reducing the scalloping loss. The periodogram amplitude at the interpolated position is

$$\begin{aligned} \text{Per}_{\mathbf{F}}(\tilde{n}, \tilde{m}) &:= \text{Per}_{\mathbf{F}}(\hat{n}, \hat{m}) - \\ &\frac{1}{4}((\text{Per}_{\mathbf{F}}(\hat{n} - 1, \hat{m}) - \text{Per}_{\mathbf{F}}(\hat{n} + 1, \hat{m}))(\tilde{n} - \hat{n}) \\ &+ (\text{Per}_{\mathbf{F}}(\hat{n}, \hat{m} - 1) - \text{Per}_{\mathbf{F}}(\hat{n}, \hat{m} + 1))(\tilde{m} - \hat{m})) \end{aligned} \quad (3.86)$$

See Appendix B for a derivation.

To use the quadratic interpolation, the dimensions of the quantized periodogram must be at least twice as large as those of $\mathbf{F}$, i.e. $N_{\text{Per}} \geq 2N$ and $M_{\text{Per}} \geq 2M$. Otherwise, one of the supporting points in (3.84) or (3.85) might actually not be on the parabola, but caused by noise or other spurs, potentially causing a large error.

Interpolation by optimization A different approach to interpolation is to re-cast the ML estimate as an optimization problem. Consider (3.47): while there is no explicit solution to this maximization problem, it could be solved by defining the periodogram with continuous arguments as an

objective function

$$\text{Per}_{\mathbf{F}}(\tilde{n}, \tilde{m}) = \frac{1}{NM} \left| \sum_{k=0}^{N_{\text{Per}}-1} \left(\sum_{l=0}^{M_{\text{Per}}-1} (\mathbf{F})_{k,l} e^{-j2\pi \frac{l\tilde{m}}{M_{\text{Per}}}} \right) e^{j2\pi \frac{k\tilde{n}}{N_{\text{Per}}}} \right|^2, \quad (3.87)$$

which is similar to the periodogram (3.30), but uses continuous values as well as a rectangular window. Without any prior knowledge of the estimates, this is not a practical optimization problem, but if a previous, coarse estimate exists, the optimization problem becomes convex and can be solved with numerical methods.

Consider Fig. 3.12, which shows both the discrete and continuous periodogram of a one-dimensional discrete-time sinusoid,

$$s(k) = e^{j2\pi \frac{\tilde{n}}{N} k}, \quad k = 0, \dots, N-1. \quad (3.88)$$

$\hat{n}$ is the coarse estimate for $\tilde{n}$, which is offset from the true value by the quantization error. Because the coarse estimate is chosen as the maximum of the discrete periodogram, it must be a point on the main lobe. Consequently, $\hat{n}$ must be the integer value that minimizes $|\hat{n} - \tilde{n}|$. Also, the double zero-padding ensures that the values $\text{Per}_{s(k)}(\hat{n}+1)$ and $\text{Per}_{s(k)}(\hat{n}-1)$ also lie on the main lobe (a rectangular window's main lobe is four samples wide for $N_{\text{Per}} = 2N$). The main lobe is convex (as $\frac{\sin(\pi x)}{\pi x}$ is convex on the interval $x \in [-1, 1]$). Therefore, estimating the true peak position of the continuous periodogram is identical to finding the maximum of a convex function on the interval $\hat{n}-1, \dots, \hat{n}+1$.

Going back to the two-dimensional periodogram, and given coarse estimates $(\hat{n}, \hat{m})$, the fractional indices are obtained by solving the optimization problem

$$\begin{aligned} (\tilde{n}, \tilde{m}) &= \arg \max_{(n,m)} \text{Per}_{\mathbf{F}}(n, m) \\ &\text{subject to} \\ \hat{n}-1 &\leq n \leq \hat{n}+1 \text{ and } \hat{m}-1 \leq m \leq \hat{m}+1 \end{aligned} \quad (3.89)$$

using the continuous periodogram from (3.87). At the same time, because the continuous periodogram is used as the objective function, the result $\text{Per}_{\mathbf{F}}(\tilde{n}, \tilde{m})$ can be used to estimate the target's RCS without scalloping loss.

One algorithm able to solve this bounded, two-dimensional optimization problem is the *Nelder-Mead simplex algorithm* [75]¹⁰ (for one dimension, [87] suggests a secant method which would not work for this two-dimensional problem). This algorithm requires initial values, which can either be the coarse indices $(\hat{n}, \hat{m})$ or even fractional indices from a previous quadratic interpolation for faster convergence.

3.3.6 Signal detection and false alarm rate

The algorithms presented in the previous section assumed a-priori knowledge; specifically, a coarse estimate of a target's position. In order to extract targets' positions from $\text{Per}_{\mathbf{F}}(n, m)$, a *target detector* must be part of the periodogram-based OFDM radar system (as depicted in Fig. 3.6). The target detector's job is to identify the peaks in the periodogram originating from valid radar targets, discriminating them from peaks originating from spurious influences. The two major sources for errors in this case are

- peaks caused by the AWGN, and
- ambiguities caused by targets' side lobes.

The latter case was already discussed in Section 3.3.4; essentially, two types of errors can arise from this:

1. Missed target: A valid target can be missed because it is identified as a side lobe of another, previously detected, target with higher backscattered power (decreased detection probability)
2. False alarm: A side lobe of a previously detected target can be identified as the main lobe of a non-existent target.

As mentioned previously, the choice of a suitable window matrix $\mathbf{W}$ can alleviate this problem to a great extent, at the cost of reduced resolution. This leaves the question of how to deal with peaks caused by AWGN.

¹⁰MATLAB implements this algorithm in the `fminsearch` function.

Constant False Alarm Rate

Most radar systems are configured for a *constant false alarm rate* (CFAR), and this solution is also chosen for the OFDM radar system. The precise definition of the CFAR varies in literature (compare [98] and [97], for instance), which is why the following definitions are introduced:

A *false alarm* is the event where the target detector decides that there is a target at a range and relative speed which did not contribute to the received matrix $\mathbf{F}_{\text{Rx}}$. The probability of a false alarm P_{FA} is the probability that, during the processing of a single frame, one or more false alarms occur when only noise was present ($\mathbf{F}_{\text{Rx}} = \mathbf{Z}$). Finally, the false alarm rate (FAR) is the expected number of detections per processing of one frame, for the case that only noise was present.

This definition differs from other definitions of false alarm probability (and FAR) in several respects:

- The time base for the false alarm rate is the duration of one frame (as a comparison, [98] suggests the number of false alarms per second for the FAR). This makes the results discussed here independent of the update rate.
- Clutter is explicitly *not* discussed in this context. The detection of an object that backscatters energy, but is not of interest for the application, thus does not count as a false alarm (an example would be the detection of a traffic sign in a vehicular context).
- Other systems, such as target tracking algorithms [138], might further process the output of the target detector (cf. Fig. 3.6), thereby possibly further reducing the false alarm rate.

In order to discriminate noise from signal power, a threshold η is introduced, and the periodogram is subjected to a hypothesis test,

$$\text{Per}_{\mathbf{F}}(n, m) \underset{H_1}{\overset{H_0}{\gtrless}} \eta, \quad (3.90)$$

where H_0 is the null hypothesis (no target is present) and H_1 is the hypothesis that a target contributes to the amplitude of the given bin.

To calculate the false alarm probability, let Z denote the (random) amplitude of any bin of $\text{Per}_{\mathbf{F}}(n, m)$ when only noise is present (due to the whiteness of the noise, Z is i.i.d. for all bins).

The probability that any single bin of the periodogram exceeds the threshold is

$$p_{\text{FA}, \text{bin}} := \Pr [z > \eta] = \int_{\eta}^{\infty} f_z(z|H_0) dz = 1 - F_Z(\eta|H_0) \quad (3.91)$$

$$= e^{-\frac{\eta}{\sigma_N^2}}, \quad (3.92)$$

where $f_Z(z|H_0)$ and $F_Z(\eta|H_0)$ are the PDF and CDF, respectively, of the random variable Z . The exponential term (3.92) is the result of Z being the magnitude-squared of AWGN with power σ_N^2 , and thus exponentially (χ_2^2) distributed.

To achieve a certain per-bin false alarm rate, solve (3.92) for η :

$$\eta = -\sigma_N^2 \ln p_{\text{FA}, \text{bin}}. \quad (3.93)$$

The optimality of this detection method is discussed in [98, Chap. 15]. In order to achieve a specific false alarm probability, note that, for a non-zero-padded periodogram, the false alarm probability is

$$p_{\text{FA}} = 1 - (1 - p_{\text{FA}, \text{bin}})^{NM}. \quad (3.94)$$

Solving this for $p_{\text{FA}, \text{bin}}$ and inserting into (3.93) yields

$$\eta = \sigma_N^2 \ln(1 - (1 - p_{\text{FA}})^{\frac{1}{NM}}). \quad (3.95)$$

If the requirement is a certain FAR, which is calculated by

$$\text{FAR} = NM p_{\text{FA}, \text{bin}}, \quad (3.96)$$

the threshold is set by

$$\eta = -\sigma_N^2 \ln \frac{\text{FAR}}{NM}. \quad (3.97)$$

The choice between (3.95) and (3.97) is determined by a trade-off: Fixing a $\text{FAR} \geq 1$ results in a lower threshold than using (3.95), and thus higher detection probability, but increases the burden on post-processing components downstream, as it also increases the number of false alarms.

When using the cropped periodograms as described in Section 3.3.2, the factor NM is replaced by $N_{\text{max}}(2M_{\text{max}} + 1)$ in (3.95) and (3.97). However, when zero-padding is used, these values are *not* increased by the interpolation factor. It is true that if there are more bins in the periodogram due to zero-padding there are potentially more bins to cause

false alarms. However, zero-padding does not add information to the periodogram, it merely reduces quantization error, and adjacent bins in zero-padded periodogram are correlated. A single peak in a non-zero-padded periodogram will thus cause multiple elements of a zero-padded periodogram to lie above the threshold, but as they are contiguous, the detection algorithm (see Section 3.3.7) will only detect this peak once.

Noise power estimation

Usually, the noise power σ_N^2 is not known at the receiver. To still be able to specify a threshold, the noise power may be estimated from the periodogram by averaging over those bins which do not contain a target. As this happens before the target detection, it is unclear which bins correspond to targets.

The solution is to rely on the correct parametrization of the OFDM radar system: As discussed in Section 3.3.2, there is a maximum index ($N_{\max}$) after which no more peaks should appear. By averaging over one or more rows beyond $N_{\max}$, a maximum likelihood estimate for σ_N^2 can be found by

$$\hat{\sigma}^2 = \frac{1}{M_{\text{Per}}K} \sum_{k=1}^K \sum_{m=0}^{M_{\text{Per}}} \text{Per}_{\mathbf{F}}(N_{\text{Per}}, m), \quad (3.98)$$

where K is the number of rows over which to average. Unless M_{Per} is very small, a value of 1 or 2 for K is sufficient.

3.3.7 Multi-target detection

The periodogram is a two-dimensional matrix which contains power levels for range and Doppler values. An application using a radar system as a sensor requires a precise list of targets, together with their individual range, Doppler and RCS.

To obtain such a list from the $\text{Per}_{\mathbf{F}(n,m)}$, a multi-target detection algorithm is necessary.¹¹

The previous section discussed how to separate noise from single targets. With multiple targets, additional problems arise, most importantly:

¹¹In long-range radar applications, this process is usually referred to as *automatic target recognition* (ATR).

- Sidelobes from strong peaks might be misidentified as separate targets.
- Weak targets close to strong targets might be overshadowed, even if they are further apart than the radar resolution.

These problems are not independent, and there is a trade-off when trying to minimize them. This becomes clear when taking a look at the window functions in Section 3.3.4: A window with a very high sidelobe attenuation might entirely mitigate the first problem, but will also decrease the radar resolution and thus amplify the possibility of overshadowing smaller targets (cf. also Fig. 3.10).

Also, computational complexity can be a problem. For a high update rate, the multi-target detector has to run very fast to allow real-time operation.

Two methods of detecting multiple targets are presented. Both methods use successive target cancellation, one coherent, the other binary.

One aspect of target recognition commonly discussed in long-range radar systems (e.g. air traffic surveillance radars) is the separation of clutter and targets. In short-range radar systems, such as vehicular radar, there really is no distinction between clutter and valid targets, since any scattering object might also be a relevant target, e.g. an automobile parked by the roadside might be clutter for one application, but relevant for another. More importantly, there is no distinction between potential clutter and targets in the periodogram. Clutter analysis is therefore not a part of this section, and will be left to other sub-systems in practice.

Binary Successive Target Cancellation

This algorithm is the simplest, both in terms of computational complexity as well as memory requirement. It relies on the assumption that a window function with a high sidelobe attenuation is used, such that sidelobes are always attenuated below the threshold.

It requires a binary map $\mathbf{B} \in \{0, 1\}^{N_{\text{Per}} \times M_{\text{Per}}}$ with the same dimensions as the periodogram, which is used to track the position of previously identified targets. It also requires the size of the window's main lobe in number of bins for range (N_{win}) and Doppler (M_{win}).

The algorithm works as follows:

1. Initialize all elements of the binary map to one, $(\mathbf{B})_{k,l} = 1$, and an index value $i = 0$.
2. Find the largest peak

$$(\hat{n}_i, \hat{m}_i) = \arg \max_{n,m} \text{Per}_{\mathbf{F}}(n, m) \quad \text{s.t.} \quad (\mathbf{B})_{n,m} = 1$$

3. If $\text{Per}_{\mathbf{F}}(\hat{n}_i, \hat{m}_i) < \eta$, stop searching.
4. Identify range, Doppler and RCS of the i -th target through a suitable interpolation algorithm from $(\hat{n}_i, \hat{m}_i)$, e.g. quadratic interpolation, and add those to the list of targets.

5. Set

$$(\mathbf{B})_{k,l} = 0 \quad \forall k, l \quad \text{s.t.} \quad \frac{(k - \hat{n}_i)^2}{(\frac{N_{\text{win}}}{2})^2} - \frac{(l - \hat{m}_i)^2}{(\frac{M_{\text{win}}}{2})^2} \leq 1$$

6. Increment i

7. Continue at 2)

Fig. 3.13 shows an example for a periodogram with three targets, the estimator result as well as $\mathbf{B}$ for iterations $i = 0$ and $i = 2$.

The major advantage of this algorithm is its simplicity. This comes at a cost: If the assumption that all sidelobes are attenuated below η is incorrect, the sidelobes *will* cause false detections. Furthermore, the window functions employed increase the resolution of the radar system.

Another disadvantage stems from the binary nature of the cancellation algorithm: When two targets lie very close together (on the order of the resolution), the peak of the weaker target may lie inside the main lobe of the stronger one. The maximum of $\text{Per}_{\mathbf{F}}(n, m)$ for which $(\mathbf{B})_{n,m} = 1$ may thus not be the true peak of the target. This can cause a target to be incorrectly identified as further away, and with a smaller RCS (see Fig. 3.13c).

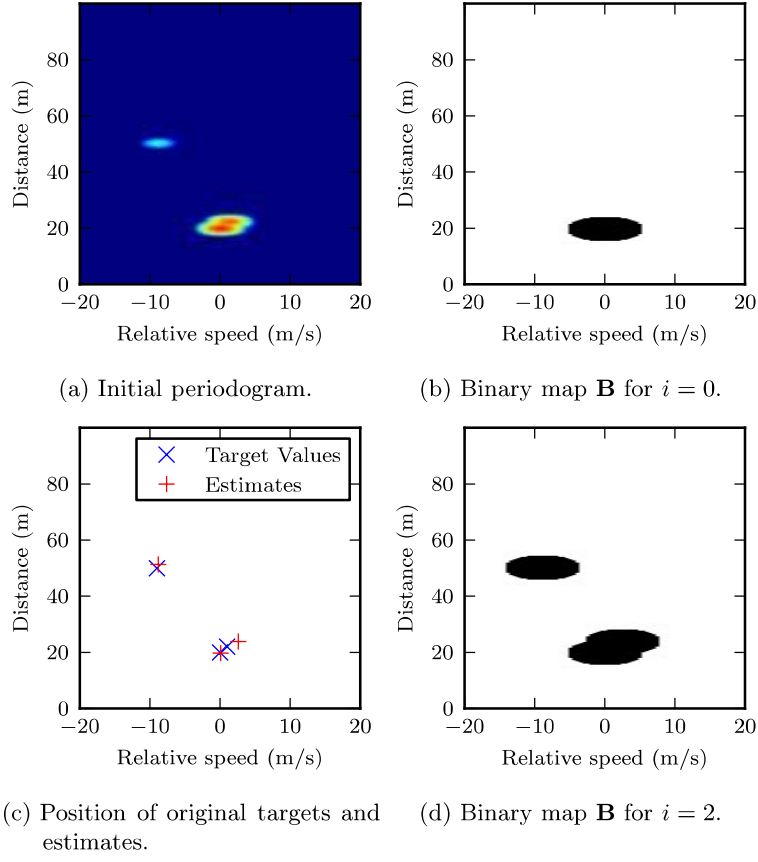


Figure 3.13: Target detection with the binary target cancellation algorithm (using a Dolph-Chebyshev window, $a_{\min} = 60$ dB)

Coherent Successive Target Cancellation

The disadvantages of the previous algorithm stem from the binary nature of the cancellation; once a bin is marked with zero in $\mathbf{B}$, it can't be used for other targets. This can be avoided by *coherently* cancelling out targets, as described in [9]. To achieve this, the algorithm must have access to both the periodogram and the complex periodogram. This is an iterative process, and as before, i shall be used to denote the current iteration, and $\text{Per}_{\mathbf{F},i}(n, m)$ shall denote the periodogram during the i -th iteration.

Unlike the previous algorithm, it successively removes targets from the periodogram, including their sidelobes. This makes it possible to use it even with a random sub-carrier selection. The algorithm consists of the following steps:

1. Initialize an iteration counter $i = 0$.

2. Find the largest peak

$$(\hat{n}_i, \hat{m}_i) = \arg \max_{n, m} \text{Per}_{\mathbf{F}}(n, m)$$

3. If $\text{Per}_{\mathbf{F}}(\hat{n}_i, \hat{m}_i) < \eta$, stop searching.

4. Check if the peak at $\text{Per}_{\mathbf{F}}(\hat{n}_i, \hat{m}_i)$ is caused by a *residual* or a target (see below).

5. If the peak describes a target, identify range, Doppler and RCS of the i -th target through a suitable interpolation algorithm from $(\hat{n}_i, \hat{m}_i)$, e.g. quadratic interpolation, and add those to the list of targets.

6. Calculate the radar processing matrix of this single target,

$$(\mathbf{F}_{\hat{n}_i, \hat{m}_i})_{k, l} = e^{j2\pi(l - \hat{m}_i) \frac{\hat{m}_i}{M_{\text{Per}}}} e^{j2\pi(k - \hat{n}_i) \frac{\hat{n}_i}{N_{\text{Per}}}}. \quad (3.99)$$

Using this, calculate the complex periodogram of $\mathbf{F}_{\hat{n}_i, \hat{m}_i}$,

$$\text{CPer}_{\mathbf{F}_{\hat{n}_i, \hat{m}_i}, i}(n, m) = \sum_{k=0}^{N_{\text{Per}}-1} \left(\sum_{l=0}^{M_{\text{Per}}-1} (\mathbf{F})_{k, l} e^{-j2\pi \frac{lm}{M_{\text{Per}}}} \right) e^{j2\pi \frac{kn}{N_{\text{Per}}}} \quad (3.100)$$

7. Subtract this from the original complex periodogram, using the complex amplitude at $(\hat{n}_i, \hat{m}_i)$:

$$\begin{aligned} \text{CPer}_{\mathbf{F},i+1}(n, m) &= \text{CPer}_{\mathbf{F},i+1}(n, m) - \\ &\quad \text{CPer}_{\mathbf{F},i+1}(\hat{n}_i, \hat{m}_i) \cdot \text{CPer}_{\mathbf{F}_{\hat{n}_i, \hat{m}_i}, i}(n, m). \end{aligned} \quad (3.101)$$

8. Update the real-valued periodogram,

$$\text{Per}_{\mathbf{F},i+1}(n, m) = \frac{1}{NM} |\text{CPer}_{\mathbf{F},i+1}(n, m)|^2. \quad (3.102)$$

9. Increment i

10. Continue at 2)

In every step, a single target is identified and its influence is removed from the periodogram, until only noise components remain.

One critical element is the identification of residuals, as mentioned in step 4). A residual may occur when the estimated position, phase or amplitude of the target's peak is incorrectly estimated, which is usually the case in a noisy environment.

Fig. 3.14 illustrates this effect for real, one-dimensional signals. The true periodogram is incorrectly estimated, with a slight offset in both amplitude as well as position. When subtracting the incorrect estimate from the true periodogram, the difference is still significant, with two significant peaks. If these peaks lie above η , they are a residual and will be detected eventually in step 2).

Even with an estimation error, the residual will always lie within the main lobe of the estimated periodogram, as the maximum-search always returns a point on the main lobe (see Section 3.3.5). A residual can therefore be identified as lying within a main lobe of a previously detected target.

To mitigate residuals, the estimation error must be minimized. For targets with a high SNR, this can be achieved by using the interpolated values $(\hat{\hat{n}}, \hat{\hat{m}})$ to calculate $\mathbf{F}_{\hat{\hat{n}}, \hat{\hat{m}}}$ and, consequently, $\text{CPer}_{\mathbf{F}_{\hat{\hat{n}}, \hat{\hat{m}}}, i}$. To calculate the latter, the interpolation algorithm must be extended to calculate an interpolated value for the amplitude and phase as well as the position, which cannot be achieved by the linear and quadratic interpolation

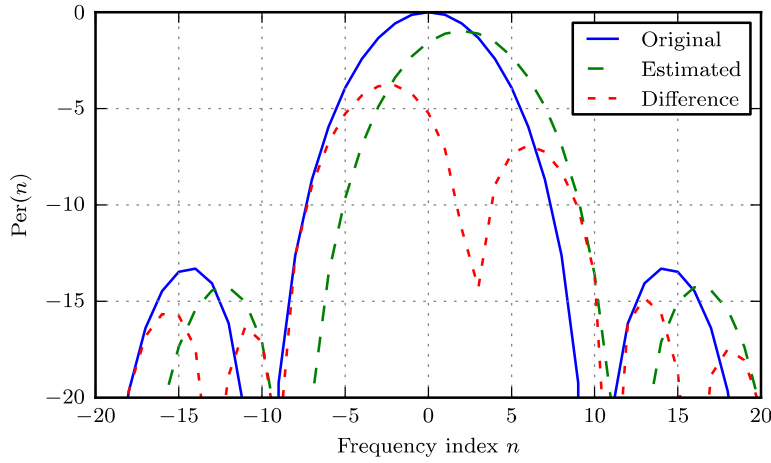


Figure 3.14: Real valued periodogram of a single target, an incorrect estimate thereof, and their difference

algorithms from Section 3.3.5 (the quadratic algorithm can interpolate the amplitude, but not the phase, whereas in the optimization algorithm, $\text{CPer}_{\hat{n}, \hat{m}, i}$ is calculated as an intermediate step in (3.89)).

The simplest way to reduce the effect of residuals is to increase the size of the periodogram $N_{\text{Per}} \times M_{\text{Per}}$ with additional zero-padding, although this increases the memory consumption and computational complexity. None of these efforts can completely remove the need to identify residuals, however, as noise will always cause an estimation error.

Because of the coherent subtraction from the periodogram between iterations, less information is destroyed per iteration when compared to the binary target cancellation. This makes the algorithm more suitable for a higher number of targets, with the increased computational effort being the only downside.

Also, as sidelobes are removed in every step, there is no requirement to use any other window than rectangular windows. In practice, a use of a window has sometimes proven beneficial as the reduced sidelobe levels prove to be more forgiving when incorrectly estimating a target in step

2).

Fig. 3.15 shows the result of the coherent cancellation algorithm for a scenario with four targets, showing some intermediary periodograms. In this simulation, the dimension of the periodogram was four times that of $\mathbf{F}$, $M_{\text{Per}} = 4M$, $N_{\text{Per}} = 4N$, and no interpolation was applied, which leads to slightly exaggerated residuals. They are correctly detected, however, and do not appear in the final list of estimated targets.

Coherent successive target cancellation is very similar to the CLEAN algorithm [59] which is often applied in radar. CLEAN factors in more problems, such as antenna positioning error, which is not relevant for the case of OFDM radar.

3.3.8 Non-contiguous sub-carrier allocations

Up until now, it was assumed that a contiguous range of N sub-carriers had been allocated. As already hinted in Section 3.1, this is only one possible way to configure an OFDM frame. This section will highlight three variants of sub-carrier allocations and discuss how they affect the radar processing.

Influence of the DC carrier

Most OFDM standards allocate a contiguous set of sub-carriers, leaving out the DC carrier, i.e. the carrier which is at 0 Hz in baseband. This avoids DC offset-related problems which typically occur in direct-conversion architectures.

This “missing” carrier causes problems for the radar component. To identify these, assume the window function for the columns $\mathbf{w}_d$ is modified such that the centre tap is set to zero (see Fig. 3.16). Since the rows of $\mathbf{F}$ are unaffected by this, it is possible to only analyze the effect in one dimension. How the DC carrier influences the OFDM radar can thus be expressed as the effect of this modified window on a sinusoid function in the periodogram.

To simplify the analysis and increase the readability, this following notation applies during this Section:

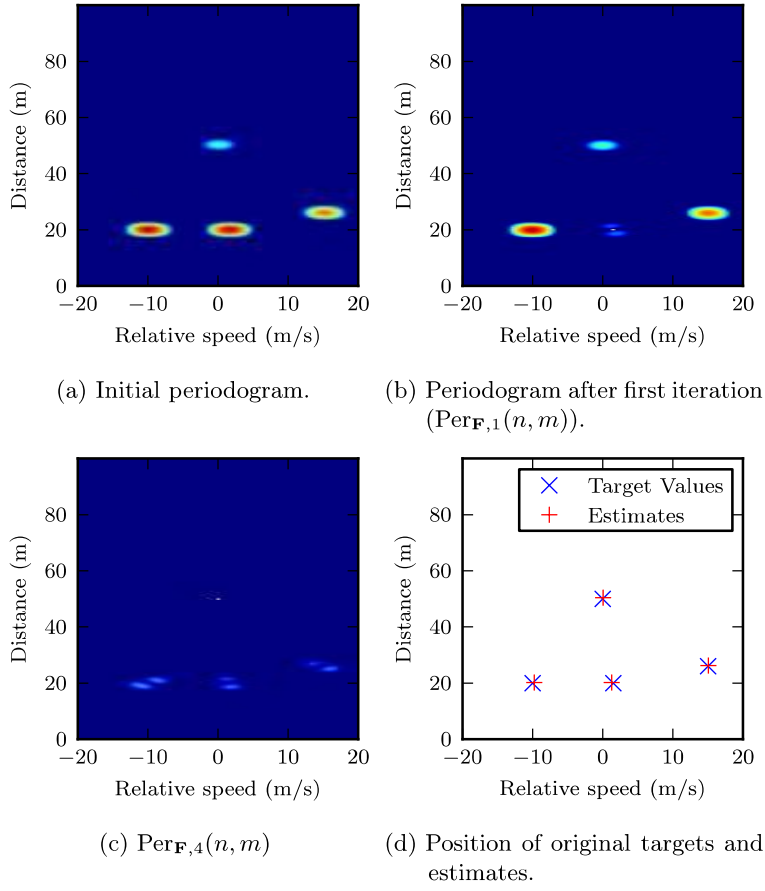


Figure 3.15: Target detection with the coherent target cancellation algorithm (using a Hamming window)

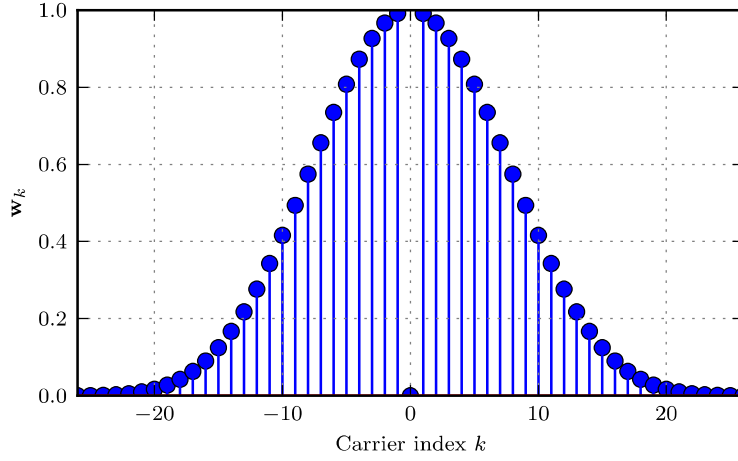


Figure 3.16: Modified Blackman-Harris window for an OFDM symbol with 52 active carriers. The centre carrier is nulled out.

- The carrier indices are rearranged such that the DC carrier has the index 0. The carrier indices therefore lie in the range $k = -\lfloor \frac{N}{2} \rfloor \dots \lfloor \frac{N}{2} \rfloor - 1$.
- The modified window is expressed as a discrete function $\tilde{w}(k)$.
- For sums, the following shorthand is defined:

$$\sum_k := \sum_{k=-\lfloor \frac{N}{2} \rfloor}^{\lfloor \frac{N}{2} \rfloor - 1}. \quad (3.103)$$

First, define the modified window function. If $w(k)$ is an unmodified window function (e.g. a Blackman-Harris window as shown in Fig. 3.16), then the modified function is given by

$$\tilde{w}(k) = w(k) - \delta(k)w(0). \quad (3.104)$$

The approximation $w(0) \approx 1$ is applied here (this is in fact an equality if N is odd for the window functions used here), which further simplifies

the modified window to

$$\tilde{w}(k) = w(k) - \delta(k). \quad (3.105)$$

The periodogram of a sinusoid with a normalised frequency $-0.5 \leq f < 0.5$ and phase φ

$$x(k) = e^{j2\pi f k} e^{j\varphi} \quad (3.106)$$

using (3.105) is thus

$$\begin{aligned} \tilde{\text{Per}}_{x(k)}(n) &= \frac{1}{N} \left| \sum_k \tilde{w}(k) x(k) e^{-j2\pi \frac{kn}{N_{\text{Per}}}} \right|^2 \\ &= \frac{1}{N} \left| \sum_k w(k) x(k) e^{-j2\pi \frac{kn}{N_{\text{Per}}}} - x(0) \right|^2. \end{aligned} \quad (3.107)$$

By using the identity $|a + b|^2 = |a|^2 + |b|^2 + 2 \text{Re}[ab^*]$, this is further modified to

$$\begin{aligned} \tilde{\text{Per}}_{x(k)}(n) &= \frac{1}{N} \left| \sum_k w(k) x(k) e^{-j2\pi \frac{kn}{N_{\text{Per}}}} \right|^2 + \frac{1}{N} |x(0)|^2 \\ &\quad - \frac{2}{N} \Re \left[x^*(0) \sum_k w(k) x(k) e^{-j2\pi \frac{kn}{N_{\text{Per}}}} \right]. \end{aligned} \quad (3.108)$$

The first summand is identical to the periodogram using the unmodified window; and $|x(0)|^2 = w(0)^2 = 1$. The third summand requires some closer scrutiny. By inserting (3.106), the term inside the real operator becomes

$$\begin{aligned} x^*(0) \sum_k w(k) x(k) e^{-j2\pi \frac{kn}{N_{\text{Per}}}} &= e^{-j\varphi} \sum_k w(k) e^{j2\pi f k} e^{j\varphi} e^{-j2\pi \frac{kn}{N_{\text{Per}}}} \\ &= \sum_k w(k) e^{-j2\pi k(\frac{kn}{N_{\text{Per}}} - f)}. \end{aligned} \quad (3.109)$$

It is identical to the discrete Fourier transform of the unmodified window, but shifted by the frequency of the sinusoid.

Finally, by taking the real part, this becomes

$$\Re \left[x^*(0) \sum_k w(k) x(k) e^{-j2\pi \frac{kn}{N_{\text{Per}}}} \right] = \sum_k w(k) \cos(2\pi k(f - \frac{kn}{N_{\text{Per}}}), \quad (3.110)$$

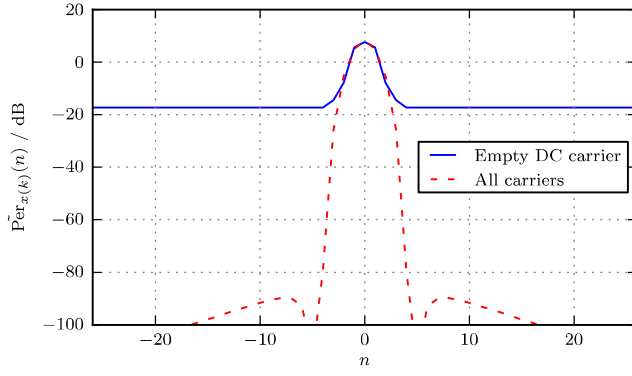


Figure 3.17: Periodogram with full set of carriers and an empty DC carrier.

and therefore the modified periodogram is

$$\begin{aligned} \tilde{\text{Per}}_{x(k)}(n) = & \underbrace{\frac{1}{N} \left| \sum_k w(k) x(k) e^{-j2\pi \frac{kn}{N_{\text{Per}}}} \right|^2}_{\text{Unmodified periodogram}} \\ & + \underbrace{\frac{1}{N} - \left(\frac{2}{N} \sum_k w(k) \cos(2\pi k(f - \frac{n}{N_{\text{Per}}})) \right)}_{\text{Spur term}}. \end{aligned} \quad (3.111)$$

Two effects are noteworthy, depending on the position in the periodogram. Near the frequency of the sinusoid, the $\cos(\cdot)$ function in the spur term is of unit value, and the spur term itself is negative:

$$\frac{1}{N} (1 - \sum_k w(k)) < 0. \quad (3.112)$$

In Fig. 3.17 it can be seen that shape of the periodogram is largely unaffected in this region, only the maximum value is reduced by a small amount. This is to be expected, since the modified window removes energy from the sinusoid. However, a far worse effect is visible for frequencies far from the sinusoid's, where a constant spur floor is evident.

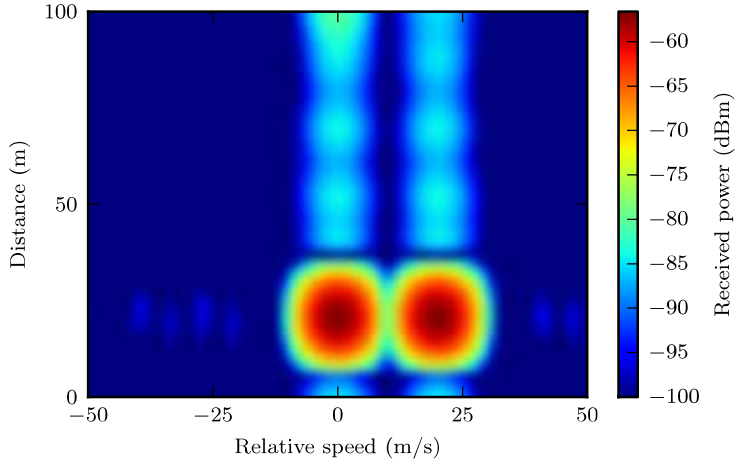


Figure 3.18: Two targets at 10 m are clearly visible; a third target at 100 m and zero Doppler can be barely distinguished from the spur floor from the closer target.

The sum in the spur term is a sum over one period of a cosine, and thus approximately zero, leaving the constant term $1/N$. As the unmodified periodogram drops to its sidelobe attenuation, the modified periodogram is kept at a minimum value of $1/N$. The peak-to-spur ratio¹² (PSR) is exactly the peak periodogram value of the modified window,

$$\text{PSR} = \left| \sum_k \tilde{w}(k) \right|^2. \quad (3.113)$$

For the radar, this means a loss of dynamic range. If two targets with the same Doppler shift are present, the difference of their respective backscattered energies cannot be larger than the PSR if both are to be detected. Fig. 3.18 illustrates this problem in the 2D periodogram. Also, the detection threshold must be adapted once a target was detected such that it is always higher than the spur floor.

¹²For a Hamming window and the carrier setup of IEEE 802.11a, PSR is approx. 27 dB. This was first observed (but not explained) by Falcone et al. [38].

To solve this problem, one solution is to simply use every second sub-carrier as in Section 3.3.8 and configure the system such that the centre carrier is left unused. If the signal parametrization is fixed (e.g. because a IEEE 802.11a system is used as a transmitter), this means ignoring half of the transmitted energy. However, losing 3 dB of dynamic range due to half the transmit power is still superior to fixing the dynamic range to the PSR.

Regular sub-carrier spacing at larger intervals

In section 3.1, it was discussed that the sub-carrier distance must be the reciprocal value of the OFDM symbol time to ensure orthogonality between carriers. Of course, this is also the case if the sub-carrier distance is an integer multiple of the inverse OFDM symbol duration, $\Delta f = U/T, U \in \mathbb{N}$.

Using a larger sub-carrier distance is equivalent to only allocating every U -th sub-carrier, leaving the others idle. Note that this does not affect the bandwidth (and hence the range resolution), but increasing Δf by a factor U will decrease the unambiguous range in (3.25) by the same factor,

$$d_{\text{unamb}} = \frac{c_0}{2U\Delta f}. \quad (3.114)$$

Also, the number of active sub-carriers is decreased which results in a higher transmit power per sub-carrier. While this does increase the signal-to-noise ratio at the receiver on each sub-carrier, the processing gain is reduced in the same manner, thereby resulting in the same SNR in the periodogram.

To summarize, a regular sub-carrier spacing only modifies the unambiguous range, the total received energy stays the same, as well as the resulting periodogram.

One reason to choose a higher sub-carrier spacing is the possibility to create multiple, orthogonal channels. If only every U -th sub-carrier is allocated, this results in U orthogonal sets of sub-carriers, which can be used by different sub-carriers. This method was first described in [18] and more elaborately in [21].

On a side note, this method can be used to avoid the problem of allocating the DC carrier by simply not using that particular set of sub-carriers.

Random sub-carrier allocations

The previous method states that up to U users can access the medium at the same time, but does not elaborate on how the users decide which sub-carrier set to transmit on. In an ad-hoc scenario, there is no guarantee that two users will not use the same channel, thereby causing mutual interference. Of course, if more than U transmissions are active, interference is guaranteed – but it is not distributed fairly among the users, as would be desirable.

A method to achieve a fair distribution of interference is to randomly select $K = N/U$ carriers for transmission, leaving the rest of them idle [11, 139]. On the radar side, such a system is simple to implement: Prior to transmission, an active carrier list $\mathbf{N}$ is generated by randomly selecting K values from the range of total carriers, $\mathbf{N} \in \{0, \dots, N-1\}^K$. The transmit matrix then has the following form:

$$(\mathbf{F}_{\text{Tx}})_{k,l} = \begin{cases} \sqrt{U}c_{k,l} & \text{if } k \in \mathbf{N} \\ 0 & \text{otherwise.} \end{cases} \quad (3.115)$$

Only the randomly selected sub-carriers are utilized, the rest are initialized with zeros (as in the previous section, the amplitude of the carriers utilized is increased by $\sqrt{U}$ to achieve the same total transmit power).

The radar processing as described in Section 3.3 is unaffected by the fact that some sub-carriers are idle, only the division (3.19) needs to be adapted to avoid a division by zero,

$$(\mathbf{F})_{k,l} = \begin{cases} \frac{(\mathbf{F}_{\text{Rx}})_{k,l}}{(\mathbf{F}_{\text{Tx}})_{k,l}} & \text{if } k \in \mathbf{N} \\ 0 & \text{otherwise.} \end{cases} \quad (3.116)$$

This affects the estimation of the number of sinusoids and their frequencies: They now have to be estimated from a set of *non-regularly sampled* data.

Spectral analysis of irregularly sampled data is a topic of its own, but a number of solutions have been found for this problem, such as the linear and non-linear least squares algorithms [91, 92, 139]. Unfortunately, these require knowledge on the number of targets, which is usually unknown in radar processing.

A method which does not have this problem and is also efficiently calculable is the periodogram. For one-dimensional signals $x(t)$ sampled at times t_k it is defined as

$$\text{Per}_{x(t_k)}(\omega_n) = \frac{1}{K} \left| \sum_{k=0}^{K-1} x(t_k) e^{-j\omega_n t_k} \right|^2. \quad (3.117)$$

If the sampling times are chosen from a regular grid, and the signal is set to zero at all other times,

$$x(k) = \begin{cases} x(t_k = kT_S) & \text{if } k \in \mathbf{N} \\ 0 & \text{otherwise,} \end{cases} \quad (3.118)$$

the discrete periodogram has nearly the same form as for the regular sampled case:

$$\text{Per}_{x(k)}(n) = \frac{1}{K} \left| \sum_{k=0}^{N-1} x(k) e^{-j2\pi \frac{kn}{N}} \right|^2, \quad (3.119)$$

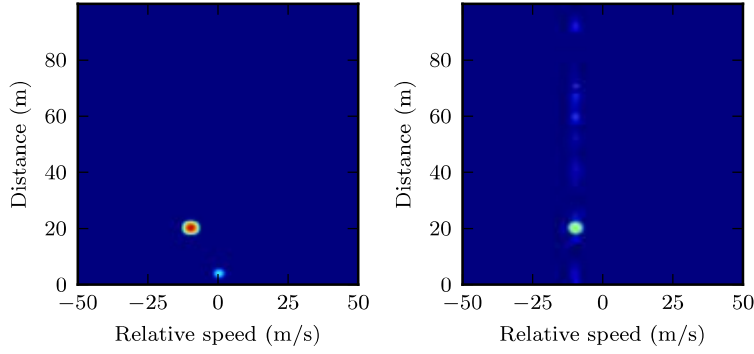
the only difference being the normalization factor.

As the radar processing matrix (3.21) represents a regular grid (the sub-carrier spacing) with the idle sub-carriers set to zero, this means that extending the periodogram to an irregular sub-carrier selection is done in the same manner as for the regular sub-carrier case,

$$\text{Per}_{\mathbf{F}}(n, m) = \frac{1}{KM} \left| \sum_{k=0}^{N_{\text{Per}}-1} \left(\sum_{l=0}^{M_{\text{Per}}-1} (\mathbf{F})_{k,l} e^{-j2\pi \frac{lm}{M_{\text{Per}}}} \right) e^{j2\pi \frac{kn}{N_{\text{Per}}}} \right|^2. \quad (3.120)$$

Most importantly, the efficient FFT-based processing is still valid.

Fig. 3.19 shows two periodograms from a simulated radar measurement using regular and irregular sub-carrier selection, respectively. One can see while the latter does not suffer from a reduction of unambiguous



(a) Regular sub-carrier spacing. An alias target can be seen at 4 m. (b) Random sub-carrier spacing. Spurs appear at the same Doppler throughout the detection range.

Figure 3.19: Scenario with a close target at 20m and a far target with large RCS at 210m.

range (there are no alias targets), it does exhibit elevated spur levels next to existing targets, similar to those in Fig. 3.18. Unlike with the missing DC carrier, the exact shape of the spur floors is random as it depends on the choice of $\mathbf{N}$. Such spurs prohibit the use of the binary target cancellation algorithm for multi-target scenarios.

Using a random sub-carrier allocation has one major advantage over the regular sub-carrier variant: If a specific sub-carrier is overlaid with strong interference, it may be discarded a-posteriori. This does decrease the total received power, but might increase the signal-to-interference ratio after the radar processing.

The disadvantages go beyond the increased spur level. On the communication side, the receiver has to detect the allocated sub-carriers, which puts additional burden on the synchronisation, equalization and demodulation.

Also, as will be discussed in Chapter 4, the interference caused by simultaneous access to the medium is not necessarily a problem for the radar system, as long as the density of the interfering transmitters is not too high.